

Final Year Project Evaluation System



Project ID: Spring-2025-04

Session: BSCS Fall 2021 to 2025

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Submitted By

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Declaration

We have read the project guidelines and understand the meaning of academic dishonesty, including plagiarism and collusion. We hereby declare that the work submitted for our final year project, entitled "Final Year Project Evaluation System," is our original work and has not been submitted, printed, or published elsewhere for any other academic or non-academic purposes.

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Statement of Submission

This is to certify that **Khazra Asif Khan** Roll No. **70129815**, **Shifa Ishaq** Roll No. **70128238** and **Afifa Kaleem Butt** Roll No. **70128235** have successfully submitted the final project titled **Final Year Project Evaluation System** at the Department of Computer Science & IT, The University of Lahore, Lahore, Pakistan, to fulfill the partial requirements for the degree of **BS in Computer Science**.

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Signature:

Date:

Dedication

This project is dedicated to our families for their unwavering support and encouragement throughout our academic journey. We especially dedicate it to our parents, whose guidance and sacrifices have been a source of inspiration, and to our mentors, who provided valuable insights to make this project a reality.

Acknowledgement

We express our deepest gratitude to our project supervisor, **Muhammad Zaman Aslam Lecturer** from **University Of Lahore**, for his invaluable guidance and constant encouragement throughout this project. We also extend our sincere thanks to the **Department of Computer Science & IT, The University of Lahore**, for providing the platform and resources needed to complete this project.

Abstract

The "Final Year Project Evaluation System" is designed to revolutionize the management of final year projects by addressing inefficiencies in traditional manual processes. It introduces a centralized, web-based platform that facilitates seamless communication, document submission, and project evaluation. This system enables students, supervisors, and evaluators to collaborate effectively through features such as a discussion forum, structured evaluation modules, and role-based access tailored to their needs.

By streamlining the workflow, the system reduces delays, improves transparency, and ensures a fair evaluation process. Students can submit their proposals, progress reports, and final documentation efficiently, while supervisors and evaluators can review, grade, and provide feedback in real-time. Additionally, the discussion forum fosters open communication, allowing students to seek clarification and guidance promptly.

The project leverages modern technologies such as React.js for the frontend and Node.js for the backend, ensuring a robust and user-friendly experience. This document provides a comprehensive overview of the system's objectives, technical specifications, and implementation strategies, demonstrating its potential to enhance academic project management significantly.

Area of the Project

We are working on a project to create a web-based platform that can be accessed on various devices, including desktops and mobile devices. The system focuses on enhancing project management through seamless communication tools, real-time feedback, and robust documentation capabilities. By integrating collaborative features and advanced evaluation methods, the platform aims to streamline the process of managing final year projects.

Technologies used

React.js: React.js is a popular JavaScript library for building user interfaces, particularly single-page applications. It enables developers to create dynamic and responsive web applications with reusable components.

- **Node.js:** Node.js is a powerful runtime environment for executing JavaScript on the server side. It facilitates efficient, scalable, and real-time applications by using an event driven, non-blocking I/O model.

- **MongoDB:** MongoDB is a NoSQL database known for its flexibility and scalability. It allows the storage and management of unstructured or semi-structured data, making it ideal for web-based applications.

- **Postman:** Postman is an API testing and collaboration tool that simplifies the process of building, testing, and documenting APIs, ensuring seamless backend integration.

- **Firebase:** Firebase is a cloud-based platform by Google that provides tools and infrastructure for app development, including authentication, hosting, and real-time databases.

- **Visual Paradigm:** Visual Paradigm is a professional UML and BPMN modeling tool. It supports creating diagrams and charts to represent system architecture, workflows, and processes effectively.

- **Word:** Word is a widely used word processing software for drafting, editing, and formatting project documentation, ensuring professional presentation and clarity.

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Chapter 1: Introduction to the Problem

1.1 Introduction

Final Year Projects (FYPs) are a vital part of the academic curriculum, offering students the opportunity to apply their knowledge to real-world challenges. Managing these projects efficiently requires collaboration between students, supervisors, and evaluators. However, traditional manual systems for overseeing FYPs often lack cohesion, making it challenging for students and faculty members to interact effectively. Communication, document submission, and feedback processes are frequently disjointed, resulting in delays and inefficiencies.

The Final Year Project Evaluation System addresses these challenges by offering a platform that not only allows for document management and evaluation but also encourages collaboration among all participants. This platform aims to streamline the process and improve communication between students, supervisors, and evaluators through tools that support discussion, documentation, evaluation and feedback.

1.2 Purpose

The purpose of this project is to develop a system that supports FYP students, supervisors, and evaluators. The platform will allow all parties to interact more effectively, upload and review documents, and provide real-time feedback. The inclusion of a discussion forum will further enhance communication. The discussion forum allows students to ask any questions related to their FYPs, and supervisors or evaluators can respond and provide clarification.

By offering a more structured and collaborative environment, this project will help fill the gaps in current systems, providing a better project management experience for students while improving faculty members' ability to guide and evaluate their work.

1.3 Objective

The objectives of the Final Year Project Evaluation System include:

- **Complete Documentation Management:** Enable students to upload project proposals, progress reports, and final submissions while allowing faculty members to review and evaluate them.
- **Facilitate Interaction:** Through the discussion forum, students can ask questions about their projects, and faculty can provide feedback on their report submissions and offer guidance.
- **Comprehensive Grading System:** Equip evaluators with a structured system to assess students' project documentation, assign grades, and provide detailed feedback, ensuring consistent and fair evaluation.
- **Role-Based Access:** Implement distinct roles for students, supervisors, evaluators and admin to ensure appropriate access to features and project data.

- **Transparency in Evaluation:** Ensure that students can track the progress of their project evaluations, receive structured feedback, and understand their strengths and areas for improvement.

1.4 Existing Solution

Through our research, we discovered that UOL's FYP platform is the primary system currently used for managing FYPs. However, it lacks several essential features that are crucial for effective project management. For instance, the platform does not include a discussion forum for students and faculty to communicate, nor does it support report submission, evaluation processes, or feedback mechanisms. These shortcomings leave students without proper tools to interact with faculty or receive timely feedback on their work. By identifying these gaps, we recognized the need for a more comprehensive solution that facilitates smoother communication and better project management for both students and faculty members.

1.5 Proposed Solution

Our proposed Final Year Project Evaluation System will address the limitations of the existing solution by providing a platform that supports both document management and interactive communication. Key features of our system include:

- **Discussion Forum:** A shared platform where students and faculty members can communicate, discuss various aspects of the project, and exchange timely feedback. This feature will facilitate open dialogue. Students can ask any questions related to their FYP's, and faculty can respond promptly to guide them.
- **Comprehensive Document Submission Support:** Students will be able to submit their documentation, such as proposals and reports, at each stage of their project. Faculty members can easily review these documents, grade them and provide feedback directly within the platform.
- **Enhanced Faculty Interaction:** Supervisors and evaluators will be able to track project progress and offer continuous feedback through the system's evaluation features.
- **Role-Based Management:** By assigning different roles to students, supervisors, and evaluators, the system ensures that everyone has access to the appropriate features based on their responsibilities.

By combining document management with a discussion forum, the Final Year Project Evaluation System will provide a complete solution that enhances communication, simplifies the evaluation process, and ensures students receive the guidance they need to succeed. This approach will make our platform more effective than existing solutions, providing a better user experience for students and faculty alike.

Chapter 2: Software Requirement Specification

2.1 Introduction

2.1.1 Purpose

The purpose of this SRS is to outline the software requirements for the development of the Final Year Project Evaluation System (FYPES). This document serves as a comprehensive guide for the development team to understand the system's functional and non-functional requirements, ensuring that all stakeholders are aligned on the project objectives and features.

Students: To understand how the system facilitates project report submission, discussion forums, and interaction with faculty.

Project Managers and Development Team: For understanding the system's specifications and ensuring proper implementation.

Supervisors and Evaluators: To ensure that the system provides the required functionalities for supervising and evaluating student projects.

Administrators: For setting up and managing system configurations and user access.

2.1.2 Scope

The Final Year Project Evaluation System (FYPES) seeks to make the management and evaluation of final-year projects easy and effective. It addresses challenges among students, supervisors, and evaluators by offering them the means of document submission, feedback, and communication.

- Students upload their project proposals, phase-1 and phase-2 chapters, and final documentation. The supervisors and evaluators are responsible for reviewing and providing feedback on these documents.
- A discussion forum for students to ask questions and get guidance from supervisors and evaluators, ensuring smooth communication.
- An announcement section where students could get notified about any update regarding FYP.
- Evaluators can easily evaluate projects based on a well-structured grading system, giving detailed feedback.
- There are different user roles, including students, supervisors, evaluators, and admins, which ensures that each user has access only to the features relevant to their responsibilities.
- Students can track the evaluation status of their project and get timely updates.

- For Students: Facilitates easy uploading of documents, asking questions, and receiving feedback that enhances the overall project experience.
- For Supervisors and Evaluators: It simplifies reviewing, evaluating, and communicating with students.
- For Administrators: It provides easy management of user roles, project data, and system configurations.

2.1.3 Definitions, acronyms, and abbreviations

Final Year Project (FYP): A study undertaken by final-year students.

Evaluator: A faculty member responsible for grading and providing feedback.

Supervisor: A faculty member guiding students through their FYP process.

Admin: A FYP head who can view and manage the activities of students, supervisors, and evaluators.

FYP: Final Year Project

SRS: Software Requirements Specification

FYPES: Final Year Project Evaluation System

UOL: The University of Lahore document.

2.2 Overall description

2.2.1 Product perspective

The FYP Evaluation System is a web-based application designed to streamline the process of final year project management for students, faculty, and administrators. The system is independent and self-contained, but it can also integrate with university systems like student databases for user verification and information retrieval. The system handles various functionalities, including report submission, evaluation, and grading, and provides communication tools like discussion forums and announcements.

Login/Signup: For user authentication.

Student Panel: Templates, report submission, and discussions.

Faculty Panel: Report grading, student supervision, and discussions.

Admin Panel: Announcements, user management, and system administration.

System Interfaces:

The system interfaces with

- University Database: To verify student, faculty, and admin details.
- Document Repository: For uploading and accessing project reports.

User Interfaces:

- Student Interface: Layout includes dashboard for report submissions, templates download, and a discussion forum.
- Faculty Interface: A dashboard for viewing student reports, grading, and supervisor list.
- Admin Interface: A dashboard for managing announcements, viewing student and faculty activities, and handling report submissions.

Hardware Interfaces:

The system is primarily web-based and can be accessed using a standard web browser. It requires no specific hardware interfaces, except for:

- Server: Adequate hardware resources must be available on the university server to support multiple users simultaneously.
- Client Devices: Students, faculty, and admins need internet-connected devices (desktops, laptops, or mobile devices) with a web browser to access the system.

Software Interfaces:

- Operating System: The system is independent of the OS, as it is web-based. It can run on any OS that supports modern web browsers (Windows).

- Browser: The system requires browsers such as Chrome.

Communications Interfaces:

Network Protocols: The system will use standard HTTP/HTTPS protocols to communicate with the server.

- Local Network: The system can be hosted within the university's local network.
- APIs: If integrated with university systems (such as student databases), secure REST APIs will be used for data exchange.

Memory:

- **Primary Memory:** The web server hosting the system must have sufficient RAM to handle concurrent user requests (students submitting reports, faculty evaluating, etc.).
- **Secondary Memory:** Storage is required for saving student project reports, faculty evaluations, and logs. The storage size depends on the number of users and the volume of project reports (which can include large files such as PDFs or presentations).

Operations

Normal Operations:

- User login/logout, report submission, grading, announcement creation, and discussion participation.
- Data will be backed up periodically to ensure no data loss (e.g., daily backups for report submissions and grading data).

Special Operations:

- **Maintenance:** The system will undergo periodic updates to add features or fix bugs, ensuring minimal downtime during academic critical periods.

Site Adaptation Requirements

The system will require initial setup for user roles (e.g., student, faculty, admin) based on the university's structure.

- Data Initialization: User data (such as student lists and project chapters) may need to be pre-loaded into the system for smooth operation.
- Customization: The system should allow for site-specific customization, such as branding with university logos or adapting features based on unique academic requirements.

2.2.2 Product functions

1. Login/Signup/Logout Modules

ID:	FR_01			
Name:	Signup (Create Account)			
Description	Input	Output	Requirements	Basic Work Flow
Allows students, faculty, and admins to create an account for accessing the system.	Name, Email, Password, Role (Student, Faculty, Admin)	Account created successfully	Internet Connectivity, Valid email	User enters required information (name, email, and password). System checks the information for validity. If valid, the account is created and saved in the database. User is notified of successful registration.

Table 01 Functional Requirement Signup

ID:	FR_02			
Name:	Login Account			
Description	Input	Output	Requirements	Basic Work Flow
Allows users to log in to their respective accounts.	Email, Password	User successfully logged in and redirected to their dashboard.	Internet Connectivity, Existing Account	User enters login credentials (email and password). System verifies the credentials. If correct, the user is logged in and taken to their specific panel (Student, Faculty, and Admin).

Table 02 Functional Requirement Login

ID:	FR_03			
Name:	Logout Account			
Description	Input	Output	Requirements	Basic Work Flow
Allows users to securely log out of their accounts	None(action triggered by user)	User is logged out and redirected to their login page.	Internet Connectivity	User clicks on the "Logout" button in their account panel. The system clears the session and redirects the user to the login page with a confirmation message of successful logout.

Table 03 Functional Requirement Logout

2. Student Panel

ID:	FR_04			
Name:	View Templates			
Description	Input	Output	Requirements	Basic Work Flow
Allows students to view and download project templates.	None (templates are pre-uploaded by the system).	List of available templates displayed.	Internet Connectivity	Student navigates to the "Templates" section. The system displays the available templates for viewing or downloading.

Table 04 Functional Requirement View Templates

ID:	FR_05			
Name:	Download Templates			
Description	Input	Output	Requirements	Basic Work Flow
Students can download a selected project template.	Selected Template	Template file downloaded.	Internet Connectivity	Student clicks on the desired template in the "Templates" section. The system allows the student to download the selected template.

Table 05 Functional Requirement Download Templates

ID:	FR_06			
Name:	Request Supervisor			
Description	Input	Output	Requirements	Basic Work Flow
Students can request a specific supervisor, and the status (pending, accepted, rejected) of their request is displayed.	Student-ID, Supervisor-ID	Request status (Pending, Accepted, and Rejected) displayed.	Internet Connectivity	Student selects a supervisor and sends a request. The system stores the request and shows the status (pending, accepted, or rejected).

Table 06 Functional Requirement Request Supervisor

3. Admin

ID:	FR_07			
Name:	Manage Faculty List			
Description	Input	Output	Requirements	Basic Work Flow
Admin can create, remove, update, and view the list of faculty members, assigning those roles (supervisor or evaluator).	Faculty Member Details (Name, Email, Role, etc.)	Updated faculty list displayed.	Admin privileges, Internet Connectivity	Admin adds, updates, or removes faculty members. The system saves the changes and displays the updated faculty list, with assigned roles.

Table 07 Functional Requirement Admin Faculty List

ID:	FR_08			
Name:	Manage Supervisor List			
Description	Input	Output	Requirements	Basic Work Flow
Admin can create, remove, update, and view the list of supervisors.	Supervisor Details (Name, Email, etc.)	Updated supervisor list displayed.	Admin privileges, Internet Connectivity	Admin adds, updates, or removes supervisors. The system saves the changes and displays the updated supervisor list.

Table 08 Functional Requirement Admin Supervisor List

ID:	FR_9			
Name:	Manage Student List			
Description	Input	Output	Requirements	Basic Work Flow
Admin can create, remove, update, and view the list of students.	Student Details (Name, Email, etc.)	Updated student list displayed.	Admin privileges, Internet Connectivity	Admin adds, updates, or removes students. The system saves the changes and displays the updated student list.

Table 9 Functional Requirement Admin Student List

ID:	FR_10			
Name:	Manage Evaluator List			
Description	Input	Output	Requirements	Basic Work Flow
Admin can create, remove, update, and view the list of evaluators.	Evaluator Details (Name, Email, etc.)	Updated evaluator list displayed.	Admin privileges, Internet Connectivity	Admin adds, updates, or removes evaluators. The system saves the changes and displays the updated evaluator list.

Table 10 Functional Requirement Admin Evaluator List

ID:	FR_11			
Name:	View Student Documentation Results			
Description	Input	Output	Requirements	Basic Work Flow
Admin can view the complete results of student documentation submissions.	Student ID (optional)	Documentation submission results displayed.	Admin privileges, Internet Connectivity	Admin navigates to the "Results" section and selects a student or views the results for all students. Documentation results are displayed.

Table 11 Functional Requirement View Student Documentation Results

ID:	FR_12			
Name:	View Student Marks			
Description	Input	Output	Requirements	Basic Work Flow
Admin can view the marks given to students by their supervisors and evaluators, displayed in a graded list format.	Student ID; Marks from supervisors and evaluators	Graded list displayed	Admin privileges, Internet Connectivity	Admin navigates to the "Marks" section and enters a Student ID or selects a student. The system retrieves and displays a graded list with marks provided by supervisors and evaluators.

Table 12 Functional Requirement View Student Marks

4. Discussion Forum

ID:	FR_13			
Name:	Post Discussion			
Description	Input	Output	Requirements	Basic Work Flow
Allows users (students, faculty) to post and reply in discussion forums.	Post Title, Description/Content	New post created.	Internet Connectivity	User enters post title and content. System saves the post to the forum. Post becomes visible to other users.

Table 13 Functional Requirement Post Discussion

5. Announcement Section

ID:	FR_14			
Name:	Create Announcement			
Description	Input	Output	Requirements	Basic Work Flow
Admins can create announcements for all users.	Announcement Title, Content	Announcement visible to all users.	Admin privileges, Internet Connectivity	Admin enters the announcement details (title, content). System saves the announcement and displays it on users' dashboards.

Table 14 Functional Requirement Create Announcement

6. Report Submission & Grading

ID:	FR_15			
Name:	Submit Report			
Description	Input	Output	Requirements	Basic Work Flow
Students can upload their reports for evaluation.	Report file (PDF, Word)	Report submitted successfully.	Internet Connectivity	Student selects and uploads their report. System stores the report in the database under the correct faculty. Student is notified of successful submission.

Table 15 Functional Requirement Submit Report

ID:	FR_16			
Name:	Grade Report			
Description	Input	Output	Requirements	Basic Work Flow
Faculty members can grade submitted student reports.	Grade, Feedback	Graded report with feedback.	Faculty privileges, Internet Connectivity	Faculty views the submitted reports. Faculty enters the grade and feedback for the report. System stores the grade and makes it available to the student.

Table 16 Functional Requirement Grade Report

7. Supervisor List and Students

ID:	FR_17			
Name:	View Supervisor List			
Description	Input	Output	Requirements	Basic Work Flow
Students can view the list of available supervisors and send requests.	None (supervisor data is pre-loaded).	List of available supervisors.	Internet Connectivity	Student navigates to the "Supervisor List" section. System displays available supervisors. Student selects a supervisor to send a request.

Table 17 Functional Requirement View Supervisor List

ID:	FR_18			
Name:	View Students of Specific Supervisor			
Description	Input	Output	Requirements	Basic Work Flow
Faculty can view students assigned to them.	Supervisor's ID (auto-assigned)	List of assigned students.	Faculty privileges, Internet Connectivity	Faculty member logs in and navigates to their dashboard. System displays a list of students assigned to that specific supervisor.

Table 18 Functional Requirement View Students of Specific Supervisor

ID:	FR_19			
Name:	View Accepted Students			
Description	Input	Output	Requirements	Basic Work Flow
Supervisors can view the list of students who have been assigned to them.	Supervisor's ID (auto-assigned)	List of accepted students displayed.	Supervisor privileges, Internet Connectivity	Supervisor logs into their panel and navigates to the "Accepted Students" section. The system retrieves and displays the list of students assigned to them.

Table 19 Functional Requirement View Accepted Students

2.2.3 User characteristics

The intended users of the FYP evaluation system are divided into three main categories:

Students

Educational Level: Undergraduate students pursuing their final year project.

Experience: Basic familiarity with using web applications, submitting reports online, and interacting with templates and discussion forums.

Technical Expertise: Moderate technical expertise; capable of using the system to upload documents, view announcements, and participate in discussions.

Faculty (Supervisors and Evaluators)

Educational Level: Faculty members with advanced degrees (Master's or PhD) in their respective fields.

Experience: Extensive experience in supervising and evaluating student projects, grading reports, and providing feedback.

Technical Expertise: Higher technical proficiency, familiar with using online platforms for managing student projects, evaluating reports, and grading.

Admins (FYP Heads)

Educational Level: Administrative personnel, typically faculty members with significant experience in managing final year project processes.

Experience: Deep understanding of the project management process, capable of overseeing the system's functionalities, managing announcements, and maintaining smooth operations.

Technical Expertise: High-level proficiency in handling administrative tasks and system management.

2.2.4 Constraints

The following constraints may affect the development and implementation of the FYP evaluation system:

- **Regulatory Policies:** The system must comply with university policies regarding data storage, student privacy, and project evaluation procedures.
- **Hardware Limitations:** The system will primarily run on university-provided servers. Users will access the system through standard web browsers on their own devices.
- **Interfaces to Other Applications:** The system may need to integrate with existing university systems, such as student databases, to retrieve or send data (e.g., student IDs, course details).

- **Parallel Operation:** The system must allow multiple users (students, faculty, and admins) to access and operate within it simultaneously.
- **Audit Functions:** Admins should be able to monitor all activities, such as report submissions, grading, and account creations, for auditing purposes.
- **Control Functions:** Admins have control over user roles, announcement creation, and access to system functionalities.
- **Higher-Order Language Requirements:** The system will be developed using React.js (JavaScript) for the front-end and Node.js for the back-end.
- **Reliability Requirements:** The system must be reliable and available during critical times, such as report submission deadlines or evaluation periods.
- **Criticality of the Application:** The application is critical to the successful completion of final year projects, so it must ensure data integrity and functionality.
- **Safety and Security Considerations:** The system must protect user data (e.g., student reports, faculty grades) and ensure secure login and authentication mechanisms.

2.2.5 Assumptions and dependencies

Assumptions:

All users have access to stable internet connections.

Users will access the system via a web browser on laptops or desktops, though some may use mobile devices.

All users (students, faculty, and admins) will be provided with unique login credentials to access the system.

The system assumes that user information (e.g., emails, project topics) is accurately provided.

Dependencies:

The system is dependent on any university-specific systems it may interface with.

Any changes in university policies related to project evaluation, privacy, or data retention may affect system requirements.

2.2.6 Apportioning of requirements

Some features of the system may be delayed for future versions or updates, such as:

Future Versions:

Real-time chat functionality: Enabling direct communication between students and supervisors.

Advanced grading analytics: Providing reports on student performance over time.

2.3 Specific requirements

2.3.1 Functional Requirement

Account Management: Users (students, faculty, and admins) can create accounts, log in, and log out.

- **Discussion Forum:** Users can post and reply to discussions within their relevant panels.
- **Template Download:** Students can access and download project templates from their panel.
- **Report Submission:** Students can upload project reports for evaluation.
- **Grading of Reports:** Faculty can evaluate and grade student reports.
- **Supervisor List:** Students can view and request supervisors from an available list.
- **Announcement Section:** Admins can create and manage announcements visible to all users.
- **View Students:** Faculty can view the list of students assigned to them.

2.3.2 Non-functional Requirements

- **Usability:**
The system should be intuitive and easy to use for students, faculty, and admins. The interface should be clear and responsive, ensuring minimal learning curve for new users.
- **Reliability:**
The system must be available during critical periods, such as report submission and evaluation deadlines.
It should be robust, ensuring that no data is lost due to system crashes or errors.
- **Performance:**
The system should handle multiple concurrent users efficiently.
Reports and grading should be processed quickly, even during peak usage times.
- **Design Constraints:**
The system must be compatible with commonly used browsers for instance Chrome. It should be responsive and usable on mobile devices for convenience.
- **Portability:**
The system will be web-based, so it should be accessible on any device with a modern web browser and an internet connection.
- **Maintainability:**
The codebase should be well-structured and modular to facilitate future updates and maintenance.

Chapter 3: Use Case Analysis

1. Login

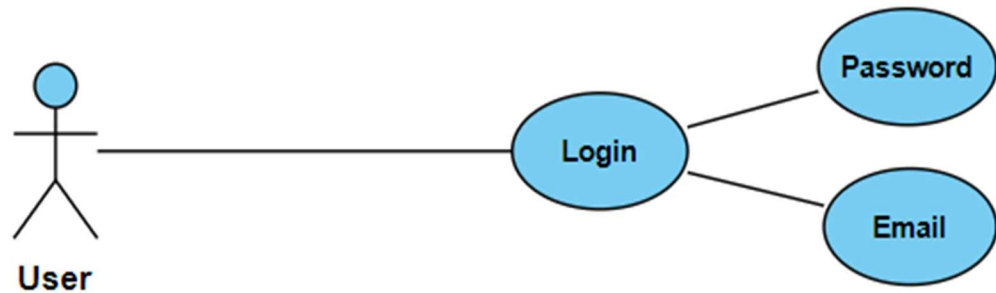


Figure 1 Use Case Diagram Login

Use Case ID	UC_01	
Use Case Name	Login	
Description	Users log into the system to access their respective portals and features.	
Primary Actor	Student, Evaluator, Supervisor, Admin	
Secondary Actor		
Pre-Condition	The user must have valid login credentials.	
Post-Condition	The user is authenticated and navigates to their dashboard.	
Basic Flow	Actor Action	System Action
	The user provides their username and password.	The system validates the credentials and displays the dashboard.
Alternate Flow	If credentials are invalid, the system prompts the user to retry the password.	

Table 10 Use Case Login

2. Sign-up:

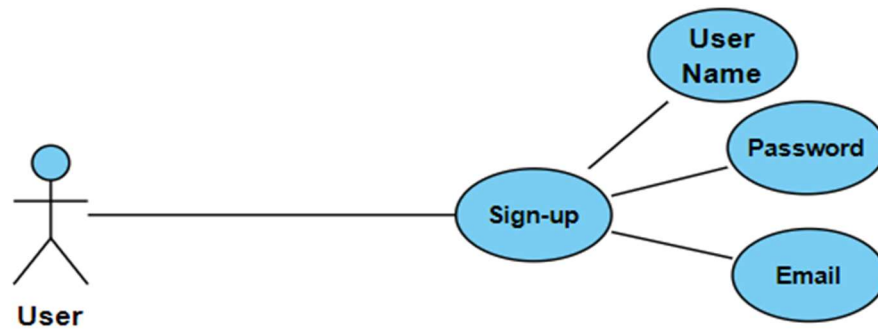


Figure 2 Use Case Diagram Sign-up

Use Case ID	UC_02	
Use Case Name	Sign-up	
Description	New students, evaluators, or supervisors create accounts to use the platform.	
Primary Actor	Student, Evaluator, Supervisor, Admin	
Secondary Actor		
Pre-Condition	The actor must provide valid registration details.	
Post-Condition	A new user account is created.	
Basic Flow	Actor Action	System Action
	The actor fills out the registration form with required details.	The system validates the input, creates an account.
Alternate Flow	If an email already exists, the system notifies the actor to use a different email address.	

Table 21 Use Case Sign-up

3. Logout:



Figure 3 Use Case Diagram Log-out

Use Case ID	UC_03	
Use Case Name	Logout	
Description	Users securely log out from the platform.	
Primary Actor	Student, Evaluator, Supervisor, Admin	
Secondary Actor		
Pre-Condition	The user is logged into the system.	
Post-Condition	The user session ends, and the user is logged out.	
Basic Flow	Actor Action	System Action
	The user clicks the "Log out" button.	The system terminates the session and redirects to the login page.
Alternate Flow		

Table 22 Use Case Log-out

4. Discussion Forum:

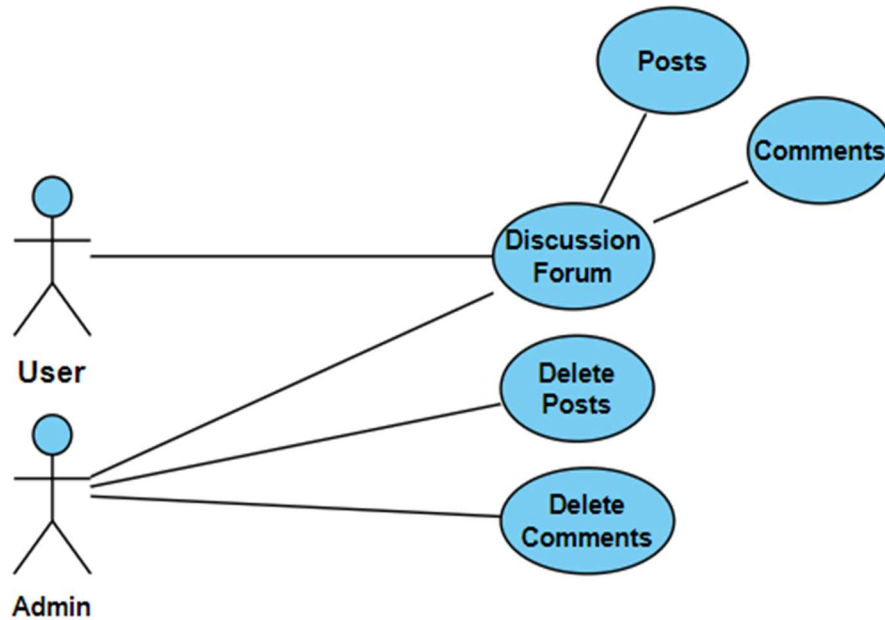


Figure 4 Use Case Diagram Discussion Forum

Use Case ID	UC_04	
Use Case Name	Discussion Forum	
Description	Students interact with other students and faculty by participating in discussions.	
Primary Actor	Student	
Secondary Actor	Admin	
Pre-Condition	The student must be logged into the system.	
Post-Condition	The forum content is updated with the student's interaction.	
Basic Flow	Actor Action	System Action
	The student posts a discussion or replies to an existing one.	The system validates and displays the input in the forum.
Alternate Flow		

Table 33 Use Case Discussion Forum

5. Announcement Section

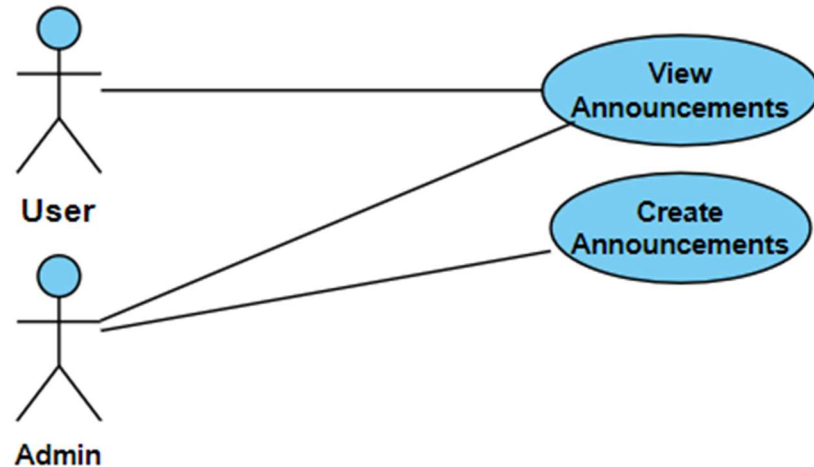


Figure 5 Use Case Diagram Announcement Section

Use Case ID	UC_05	
Use Case Name	Announcement Section	
Description	Users can view announcements created by admin.	
Primary Actor	Student Supervisor, Evaluator	
Secondary Actor	Admin	
Pre-Condition	The admin actor must have permissions to create announcements.	
Post-Condition	The announcement is successfully published.	
Basic Flow	Actor Action	System Action
	The admin actor enters the announcement details, and user actors can only view them.	The system publishes the announcement for users so they can view.
Alternate Flow	If permissions are insufficient, the system denies access and notifies the admin actor.	

Table 44 Use Case Announcement Section

6. Template

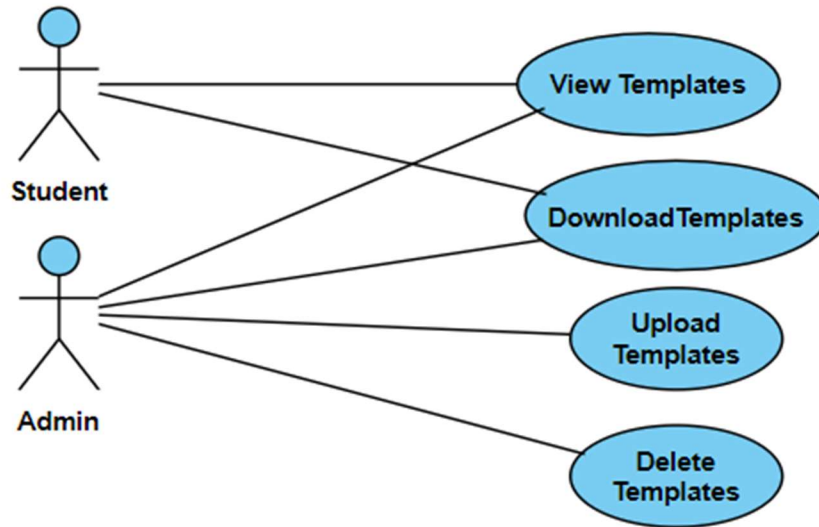


Figure 6 Use Case Diagram Template

Use Case ID	UC_06	
Use Case Name	Template	
Description	Students, evaluators, or supervisors access and use templates uploaded by admin for documentation or submissions.	
Primary Actor	Student, Supervisor, Evaluator	
Secondary Actor	Admin	
Pre-Condition	The user must be logged in.	
Post-Condition	The requested template is downloaded or accessed.	
Basic Flow	Actor Action	System Action
	The user navigates to the "Templates" section and selects a template that are uploaded by admin actor. Admin actor can also delete any template.	The system retrieves and displays or allows download of the selected template.
Alternate Flow	If the template is unavailable, the system notifies the user.	

Table 55 Use Case Template

7. Request To Supervisor

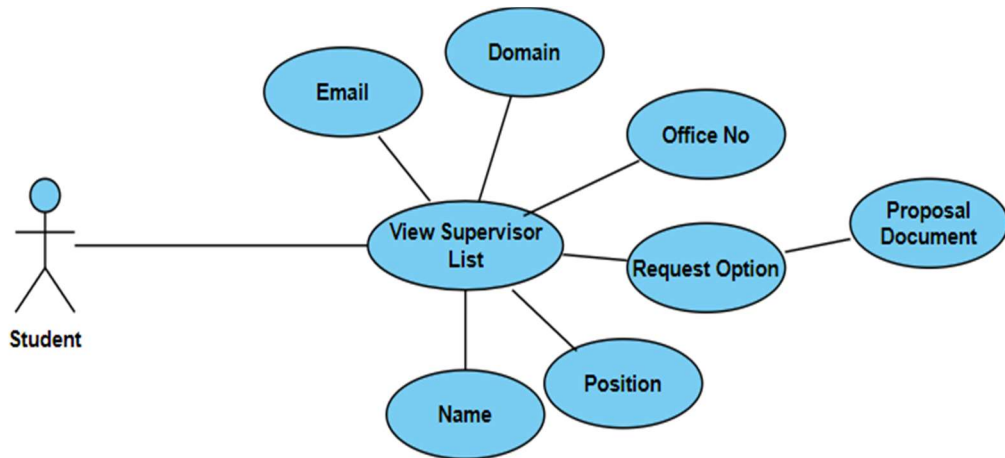


Figure 7 Use Case Diagram Request To Supervisor

Use Case ID	UC_07	
Use Case Name	Request to Supervisor	
Description	Students send requests to supervisor after viewing supervisor list.	
Primary Actor	Student	
Secondary Actor	Supervisor	
Pre-Condition	The student must be logged in.	
Post-Condition	The request is submitted to the supervisor.	
Basic Flow	Actor Action	System Action
	The student submits a request with proposal document.	The system forwards the request to the selected supervisor.
Alternate Flow	If the supervisor is accept or reject the request, student will be notified by the updated status.	

Table 66 Use Case Request To Supervisor

8. Document Submission

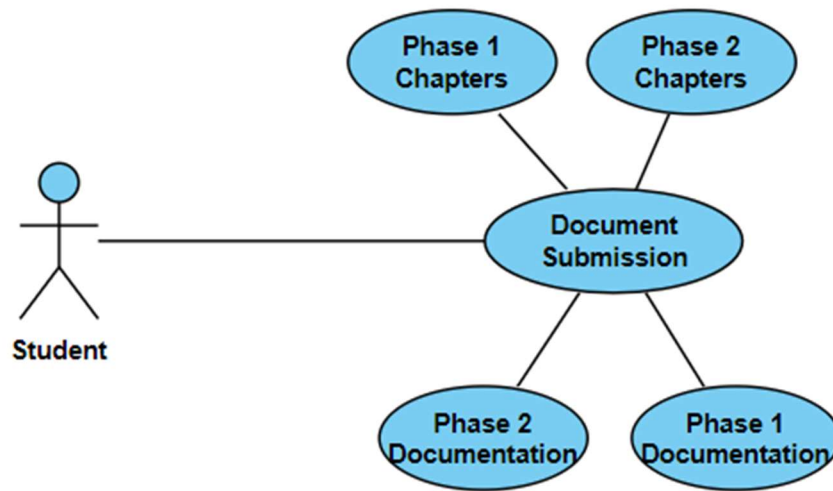


Figure 8 Use Case Diagram Document Submission

Use Case ID	UC_08	
Use Case Name	Document Submission	
Description	Students submit their documents or chapters for grading.	
Primary Actor	Student	
Secondary Actor	Supervisor, Evaluator	
Pre-Condition	The student must be logged in.	
Post-Condition	The document is submitted successfully.	
Basic Flow	Actor Action	System Action
	The student uploads the document or chapter, and submits it.	The system validates the upload, stores it, and notifies the supervisor or evaluator.
Alternate Flow	If the file format is invalid, the system rejects the upload and notifies the student.	

Table 77 Use Case Document Submission

9. View Requests By Students

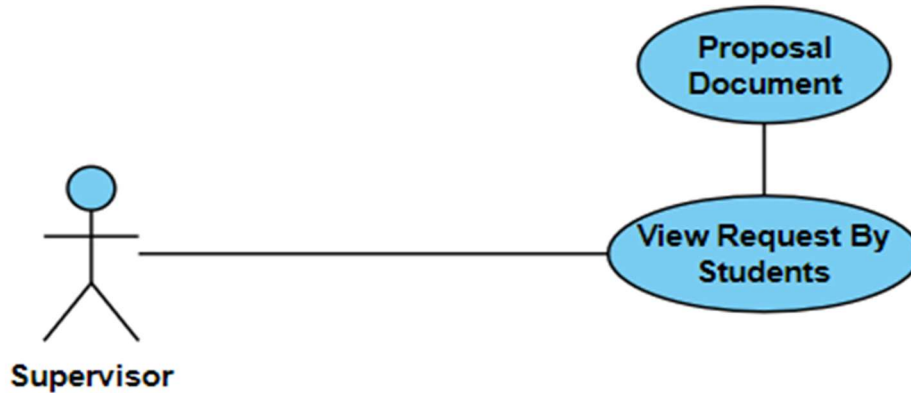


Figure 9 Use Case Diagram View Requests by Students

Use Case ID	UC_09	
Use Case Name	View Requests by Students	
Description	Supervisors view and manage requests submitted by students with proposal document.	
Primary Actor	Supervisor	
Secondary Actor	Student	
Pre-Condition	The supervisor must be logged in.	
Post-Condition	The supervisor reviews the student requests.	
Basic Flow	Actor Action	System Action
	The supervisor navigates to the "Requests" section.	The system retrieves and displays pending requests.
Alternate Flow	If no requests are available, the system displays a "No Requests Found" message.	

Table 88 Use Case View Requests By Students

10. Accepted Students List

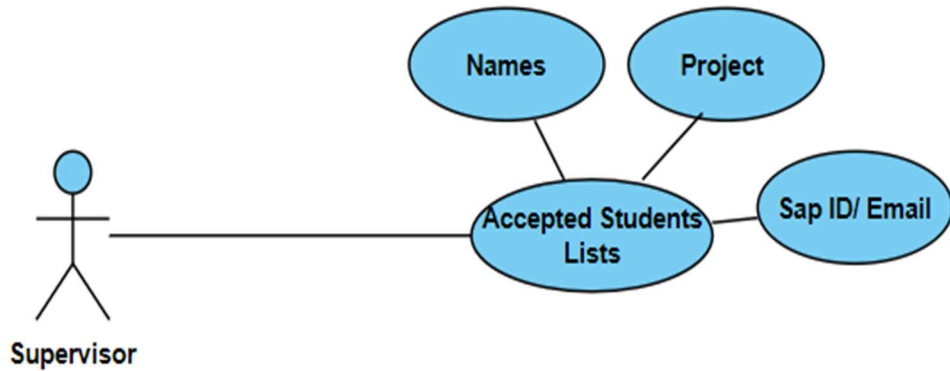


Figure 10 Use Case Diagram Accepted Students List

Use Case ID	UC_10	
Use Case Name	Accepted Students List	
Description	Supervisors maintain and view a list of accepted students under their guidance.	
Primary Actor	Supervisor	
Secondary Actor	Student	
Pre-Condition	The supervisor must have accepted students list.	
Post-Condition	The list of accepted students is updated or displayed.	
Basic Flow	Actor Action	System Action
	The supervisor accesses the "Accepted Students List."	The system retrieves and displays the updated list.
Alternate Flow		

Table 29 Use Case Accepted Students List

11. Chapters Evaluation

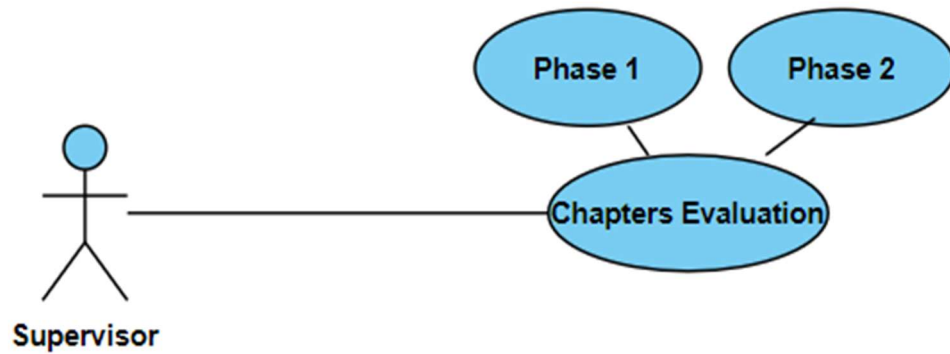


Figure 11 Use Case Diagram Chapters Evaluation

Use Case ID	UC_11	
Use Case Name	Chapters Evaluation	
Description	Supervisors evaluate submitted chapters.	
Primary Actor	Supervisor	
Secondary Actor	Student	
Pre-Condition	The student must have submitted the chapter for evaluation.	
Post-Condition	The chapter is evaluated and feedback is provided.	
Basic Flow	Actor Action	System Action
	The supervisor selects a chapter to review.	The system retrieves the chapter, records the evaluation, and sends feedback to the student.
Alternate Flow	If the chapter format is invalid, the system rejects the upload and notifies the student.	

Table 30 Use Case Chapters Evaluation

12. Document Evaluation

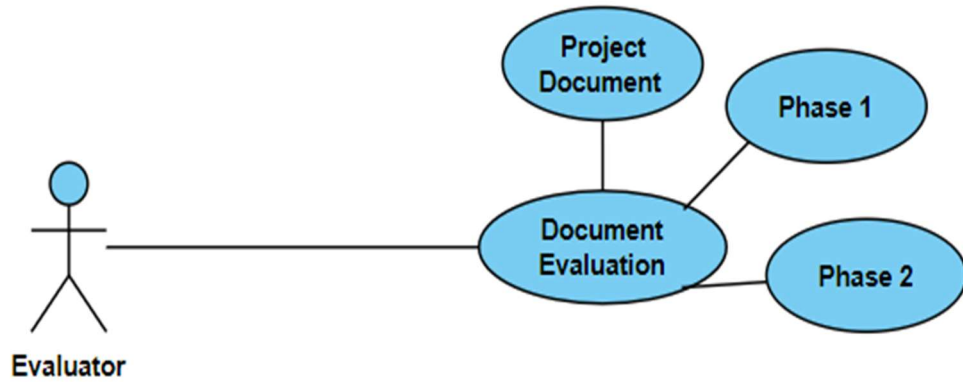


Figure 12 Use Case Diagram Document Evaluation

Use Case ID	UC_12	
Use Case Name	Document Evaluation	
Description	Evaluators evaluate submitted Document.	
Primary Actor	Evaluator	
Secondary Actor	Student	
Pre-Condition	The student must have submitted the document for evaluation.	
Post-Condition	The document is evaluated and feedback is provided.	
Basic Flow	Actor Action	System Action
	The evaluator selects a document to review.	The system retrieves the document, records the evaluation, and sends feedback to the student.
Alternate Flow	If the document format is invalid, the system rejects the upload and notifies the student.	

Table 31 Use Case Document Evaluation

13. View Assigned Students List

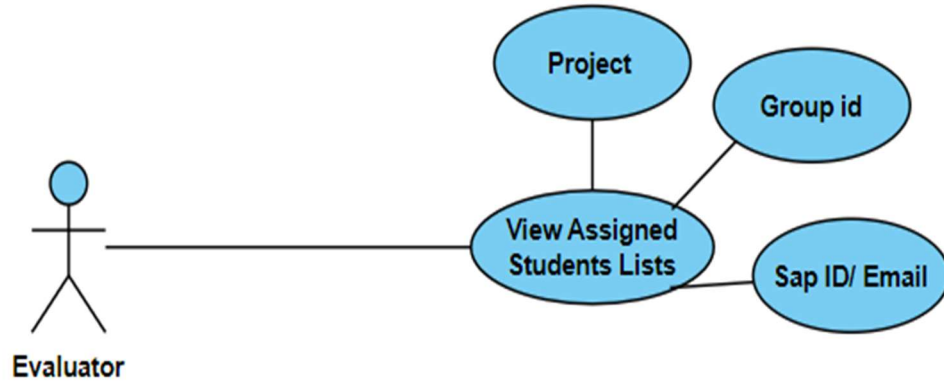


Figure 13 Use Case Diagram View Assigned Students List

Use Case ID	UC_13	
Use Case Name	View Assigned Students List	
Description	Evaluators view their assigned students list.	
Primary Actor	Evaluator	
Secondary Actor	Student	
Pre-Condition	The evaluator must have assigned students in the system.	
Post-Condition	The list of assigned students is displayed.	
Basic Flow	Actor Action	System Action
	The evaluator navigates to the "Assigned Students List."	The system retrieves and displays the list.
Alternate Flow	If no students are assigned, the system displays a "No Students Found" message.	

Table 32 Use Case View Assigned Students List

14. Faculty List

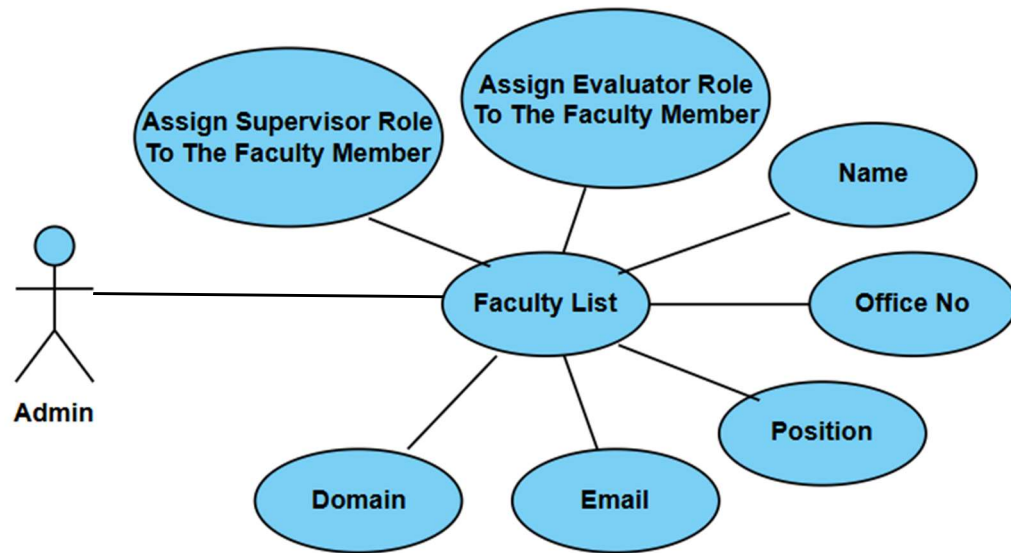


Figure 14 Use Case Diagram Faculty List

Use Case ID	UC_14	
Use Case Name	Faculty List	
Description	Admin views the list of faculty members and assigns them roles as Supervisor or Evaluator.	
Primary Actor	Admin	
Secondary Actor		
Pre-Condition	The admin must be logged in.	
Post-Condition	A faculty member is successfully assigned the role of Supervisor or Evaluator.	
Basic Flow	Actor Action	System Action
	The admin navigates to the "Faculty List" section. The admin selects a faculty member and assigns them the role of Supervisor or Evaluator.	The system retrieves and displays the list of all faculty members. The system updates the faculty member's profile with the selected role and confirms the assignment.
Alternate Flow		

Table 33 Use Case Faculty List

15. Student List

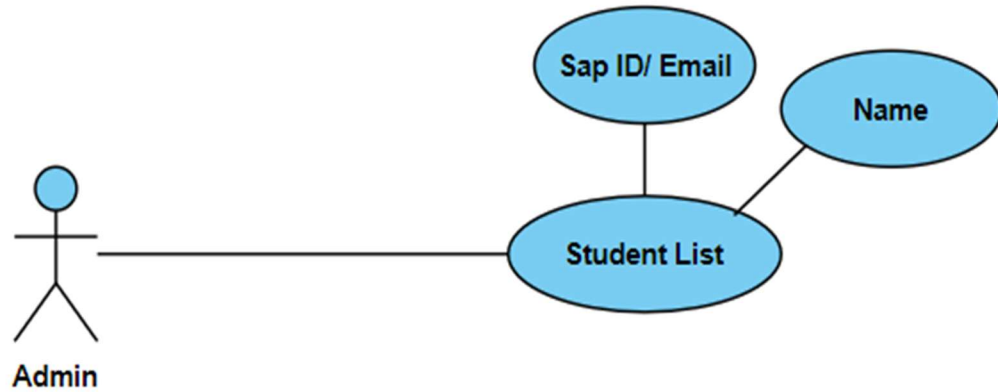


Figure 15 Use Case Diagram Student List

Use Case ID	UC_15	
Use Case Name	Student List	
Description	Admins view and manage the list of all registered students.	
Primary Actor	Admin	
Secondary Actor		
Pre-Condition	The admin must be logged in.	
Post-Condition	The list of students is displayed or modified.	
Basic Flow	Actor Action	System Action
	The admin navigates to the "Student List" section.	The system retrieves and displays the list of registered students.
Alternate Flow	If no students are registered, the system displays a "No Students Found" message.	

Table 34 Use Case Student List

16. Supervisor List

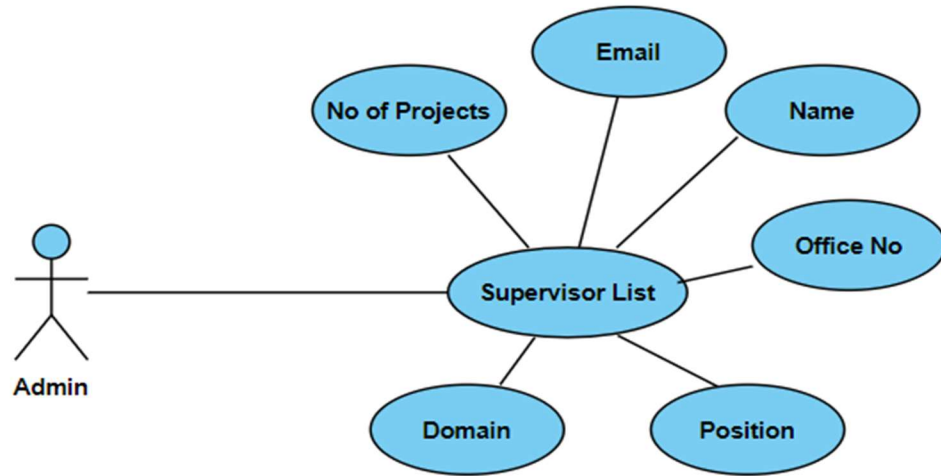


Figure 16 Use Case Diagram Supervisor List

Use Case ID	UC_16	
Use Case Name	Supervisor List	
Description	Admins view and manage the list of all supervisors.	
Primary Actor	Admin	
Secondary Actor		
Pre-Condition	The admin must be logged in.	
Post-Condition	The list of supervisors is displayed or modified.	
Basic Flow	Actor Action	System Action
	The admin navigates to the "Supervisor List" section.	The system retrieves and displays the list of supervisors.
Alternate Flow	If no supervisors are registered, the system displays a "No Supervisors Found" message.	

Table 35 Use Case Supervisor List

17. Evaluator List:

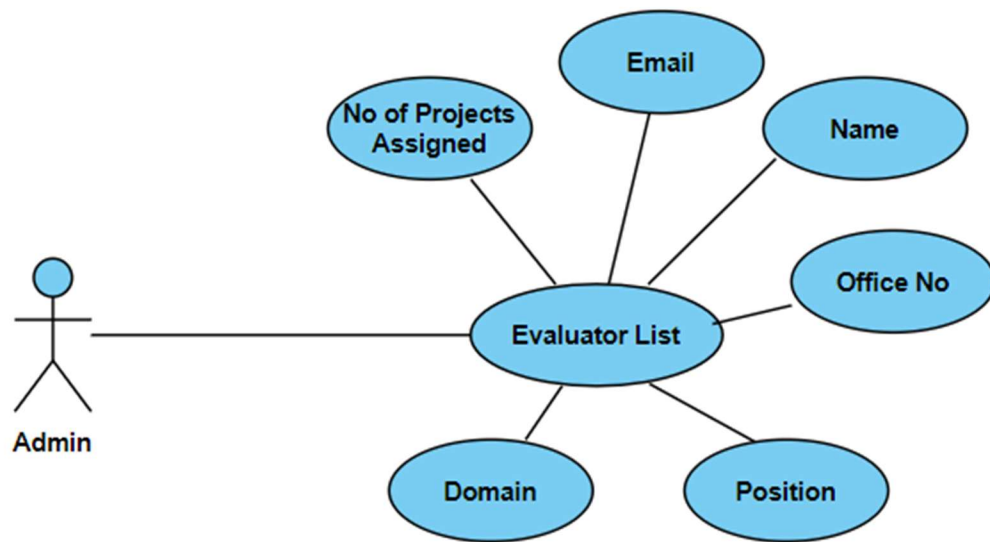


Figure 17 Use Case Diagram Evaluator List

Use Case ID	UC_17	
Use Case Name	Evaluator List	
Description	Admins view and manage the list of all evaluators.	
Primary Actor	Admin	
Secondary Actor		
Pre-Condition	The admin must be logged in.	
Post-Condition	The list of evaluators is displayed or modified.	
Basic Flow	Actor Action	System Action
	The admin navigates to the "Evaluator List" section.	The system retrieves and displays the list of evaluators.
Alternate Flow	If no evaluators are registered, the system displays a "No Evaluators Found" message.	

Table 36 Use Case Evaluator List

18. Projects Evaluation:

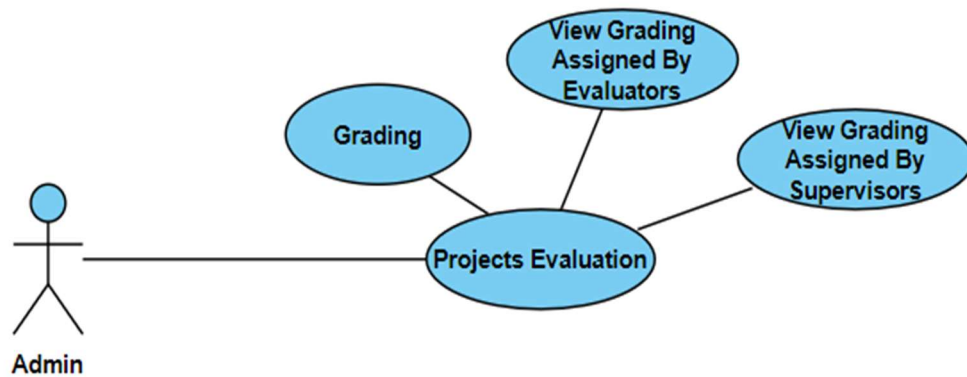


Figure 18 Use Case Diagram Projects Evaluation

Use Case ID	UC_18	
Use Case Name	Projects Evaluation	
Description	Admin assess and provide feedback on student projects.	
Primary Actor	Admin	
Secondary Actor	Student	
Pre-Condition	Students must have submitted their projects.	
Post-Condition	The evaluation results are displayed.	
Basic Flow	Actor Action	System Action
	The admin selects a project and evaluates it.	The system saves the evaluation and notifies the student.
Alternate Flow		

Table 37 Use Case Projects Evaluation

Chapter 4: Design

4.1 Architecture Diagram

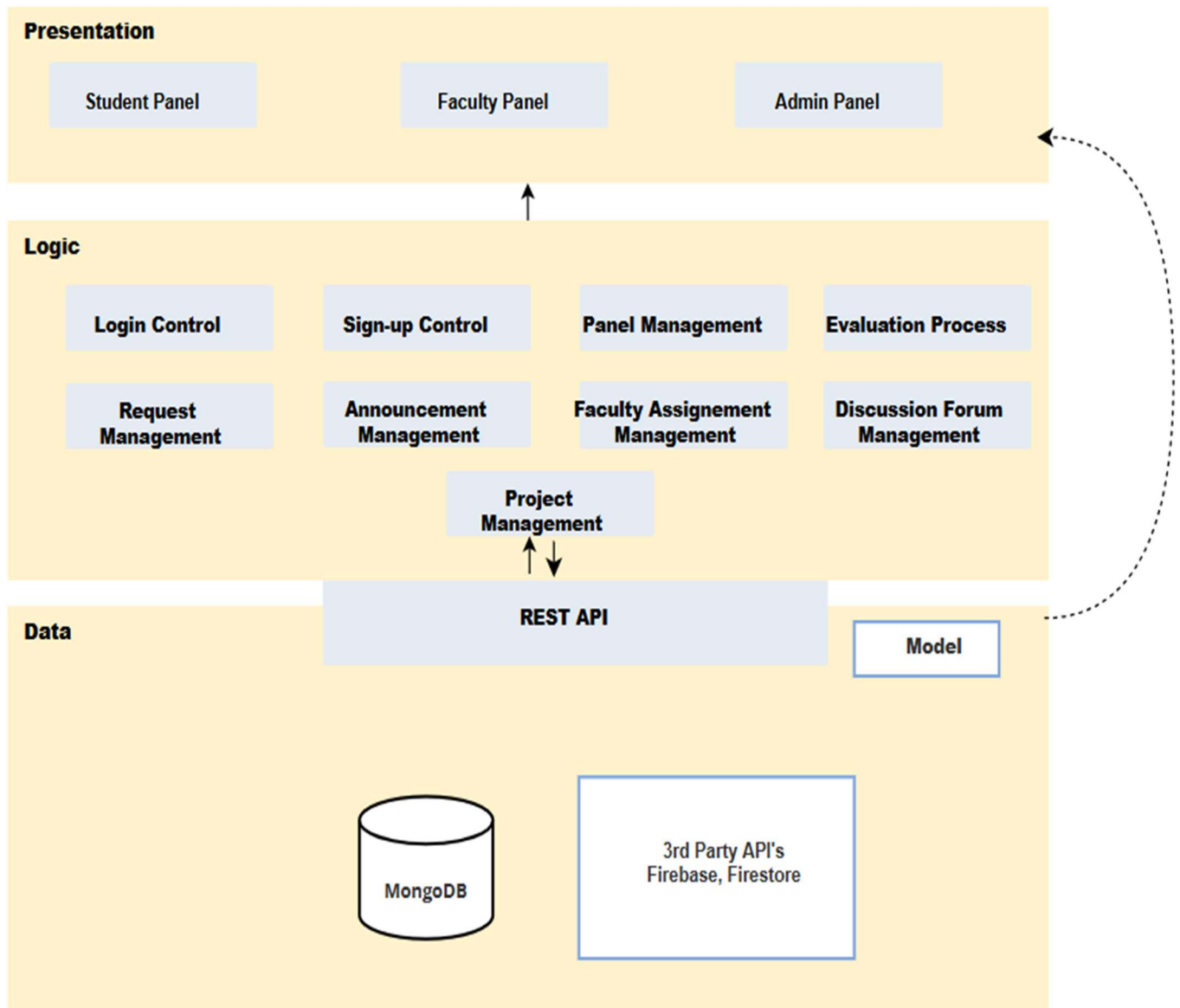


Figure 19 Architecture Diagram

4.2 ERD with data dictionary

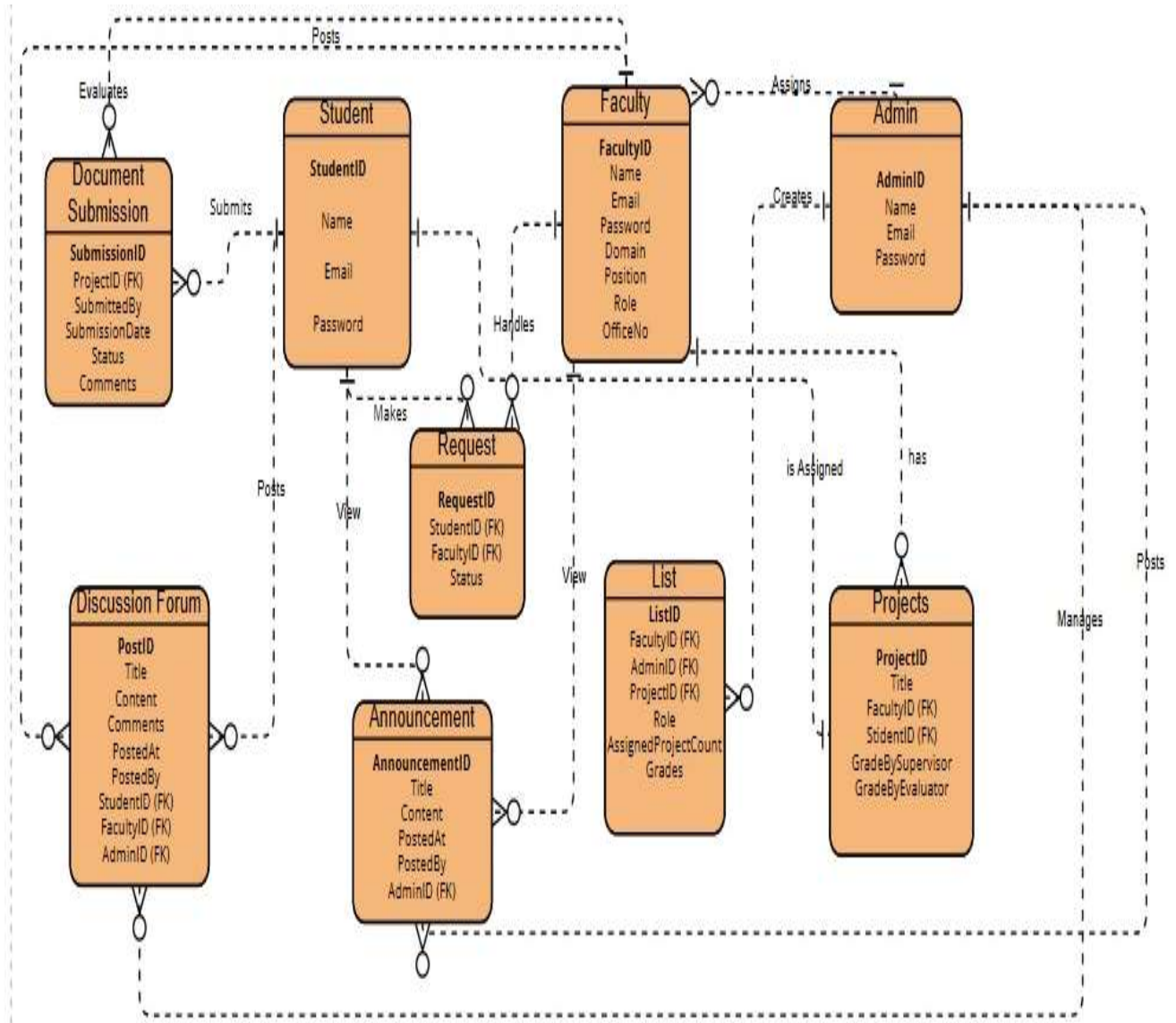


Figure 20 ERD With Data Dictionary

4.3 Data Flow Diagram

4.3.1 The Level 0

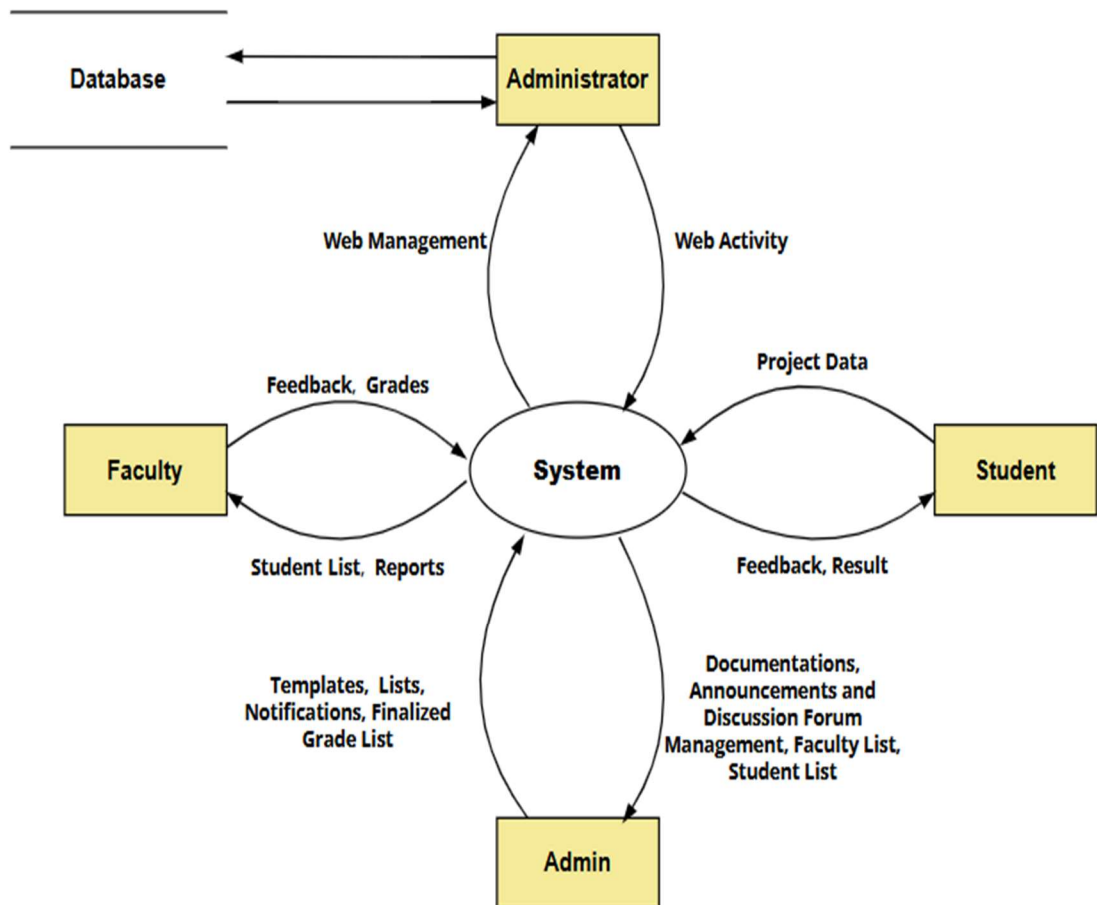


Figure 21 Level 0 DFD

4.3.2 The Level 1

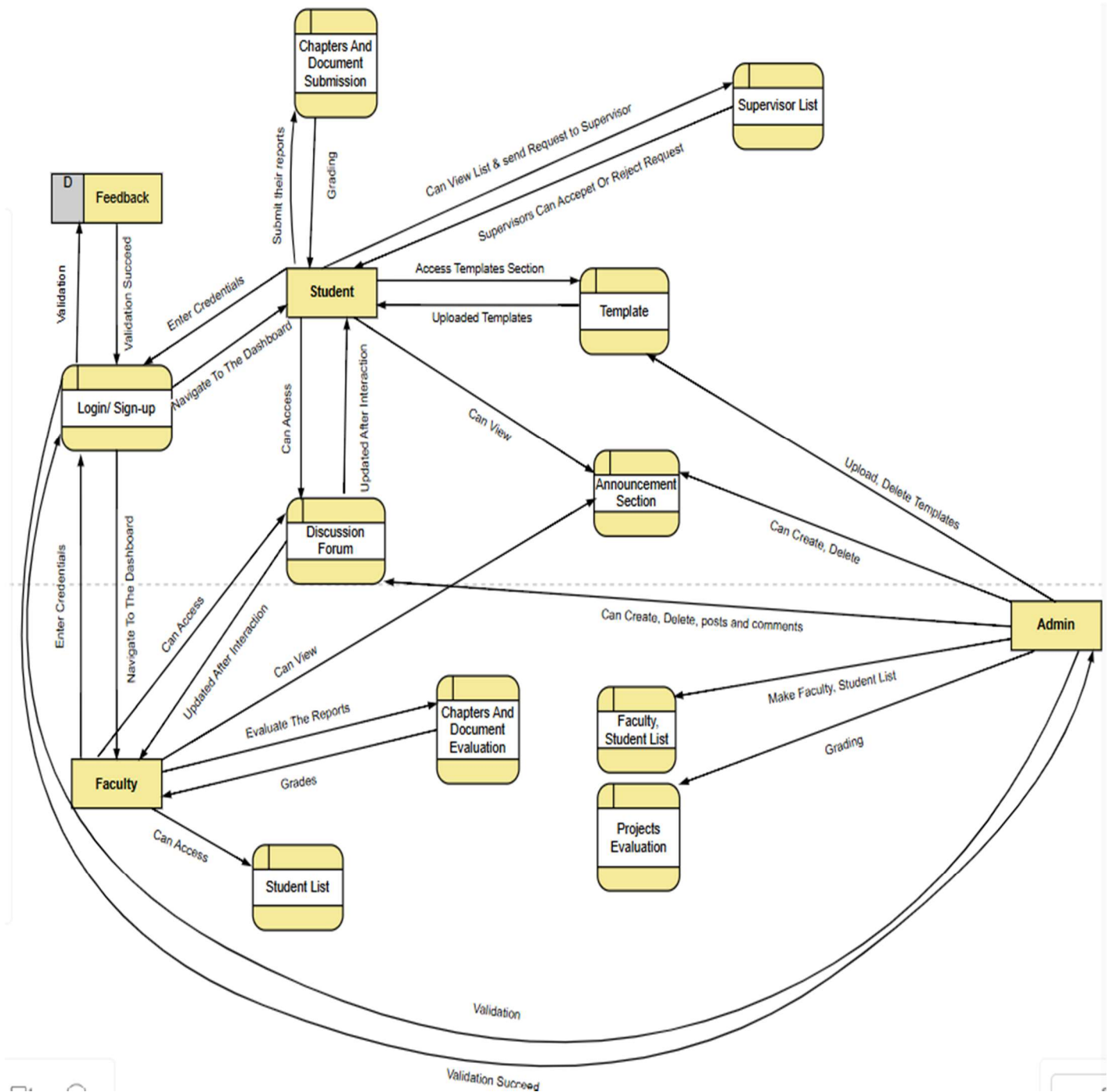


Figure 22 Level 1 DFD

4.4 Class Diagram

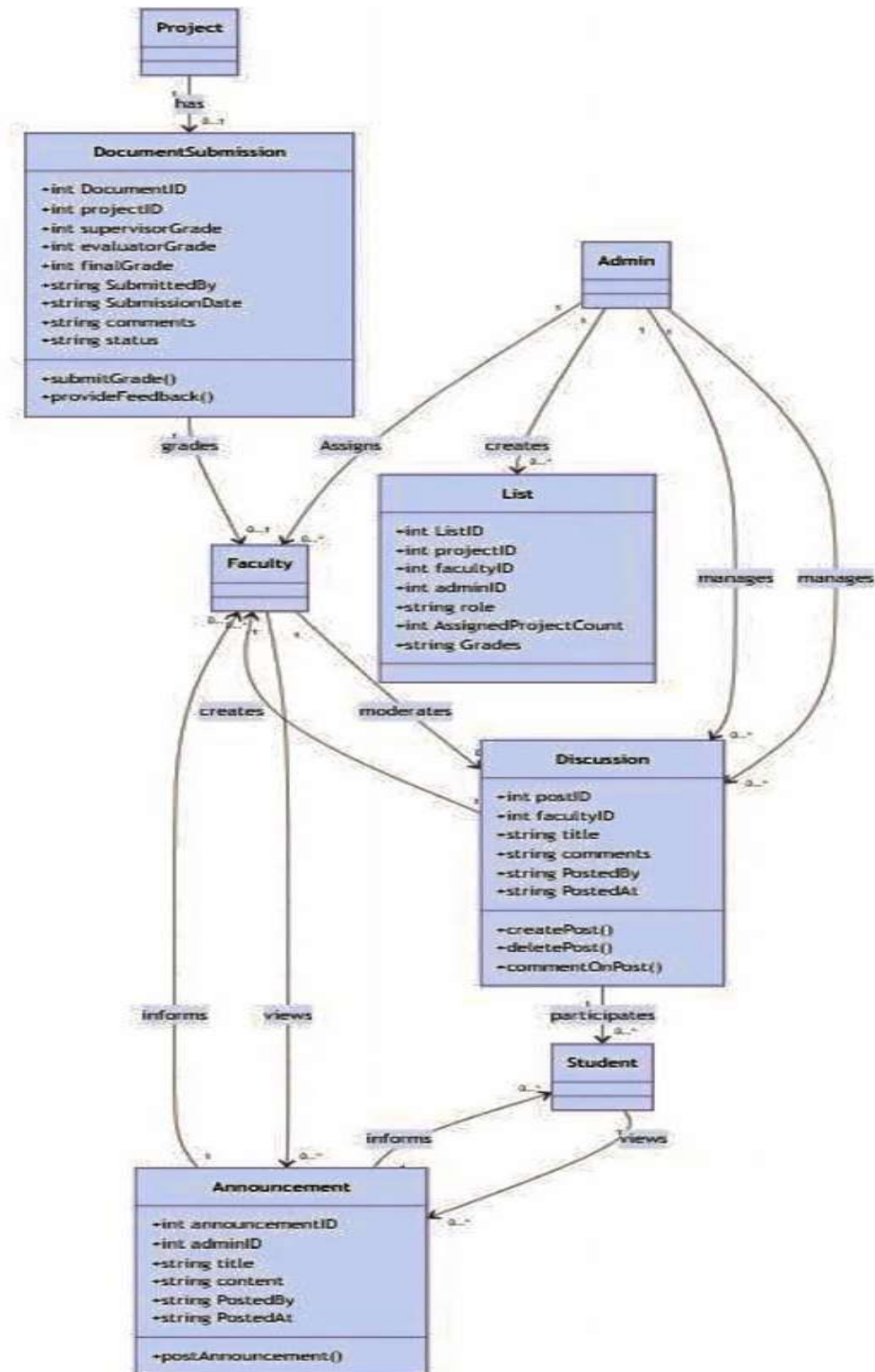


Figure 23 Class Diagram

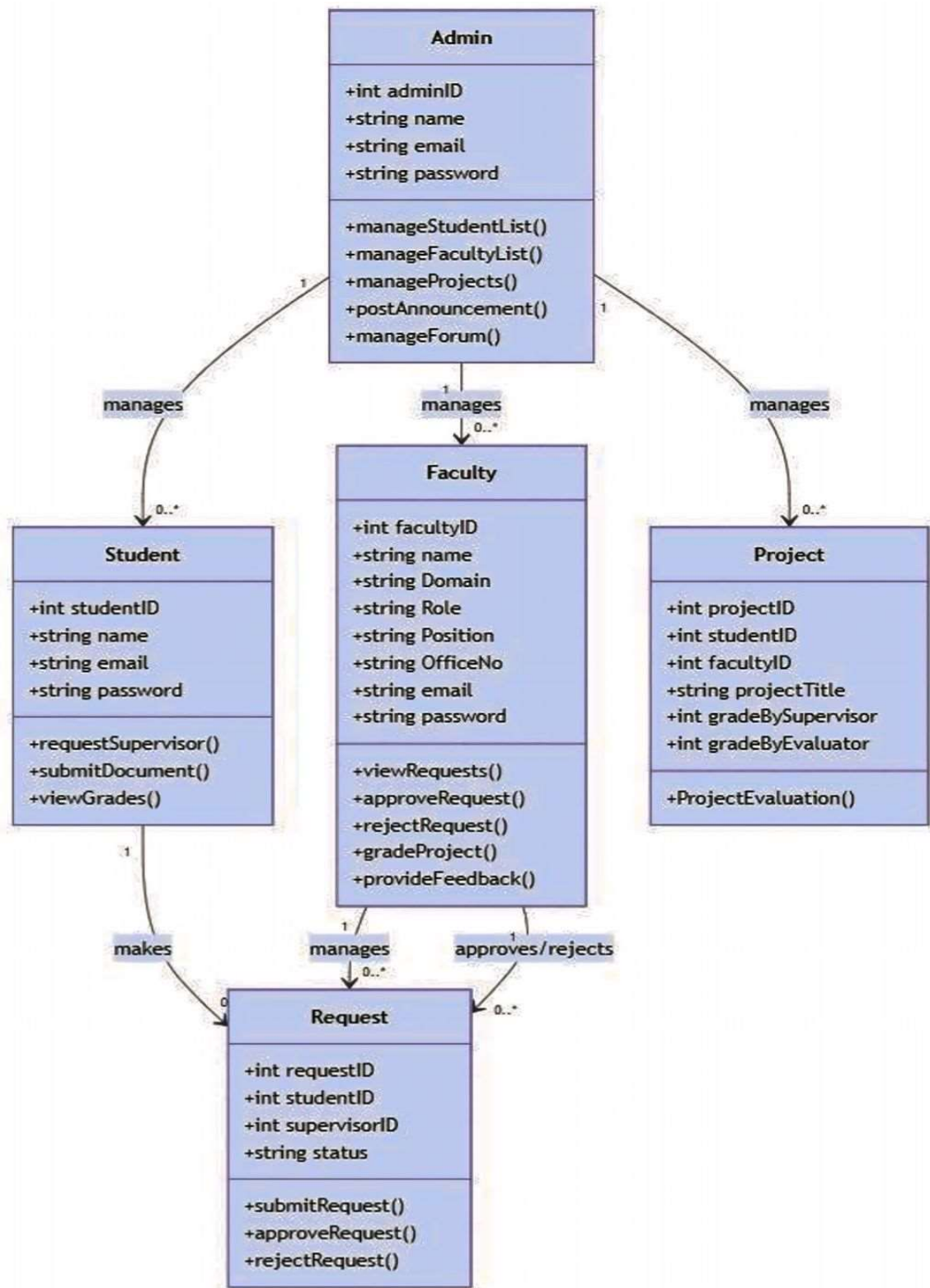


Figure 24 Class Diagram

4.5 Activity Diagram

4.5.1 Login

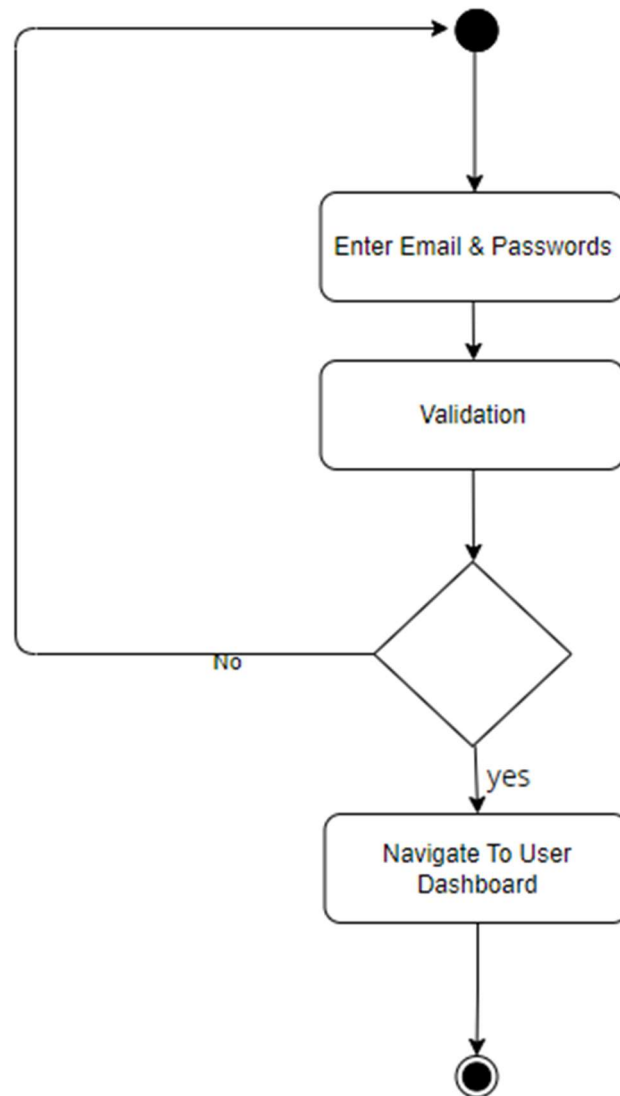


Figure 25 Activity Diagram Login

4.5.2 Logout

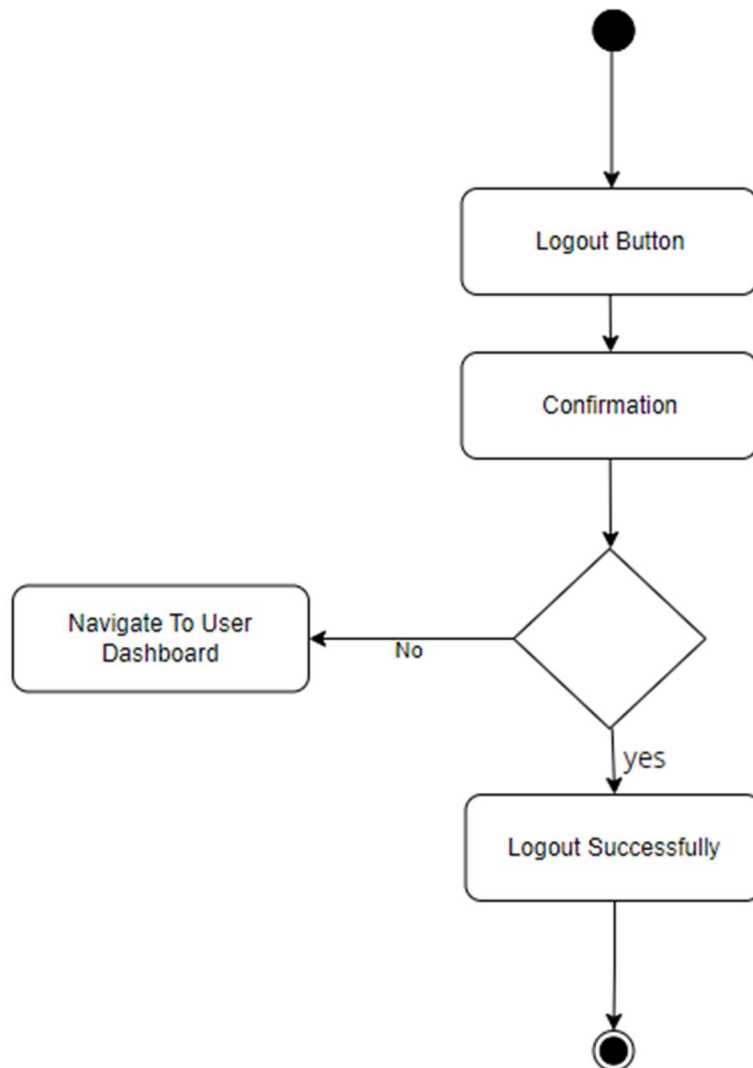


Figure 26 Activity Diagram Logout

4.5.3 Sign up

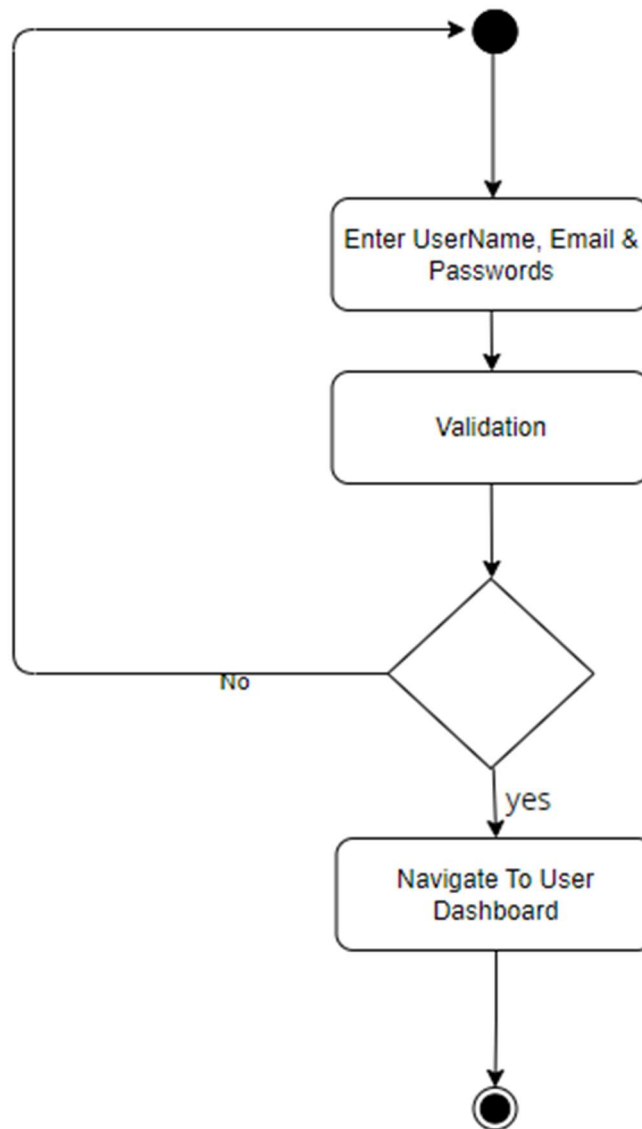


Figure 27 Activity Diagram Sign Up

4.5.4 Student Send Request To Supervisor

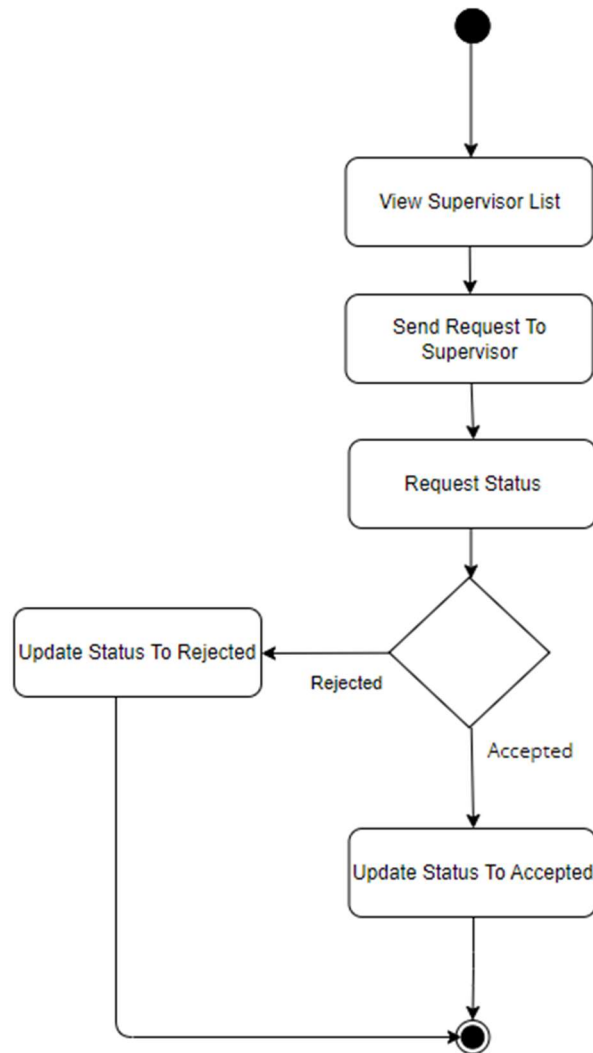


Figure 28 Activity Diagram Student Send Request To Supervisor

4.5.5 Faculty Profile

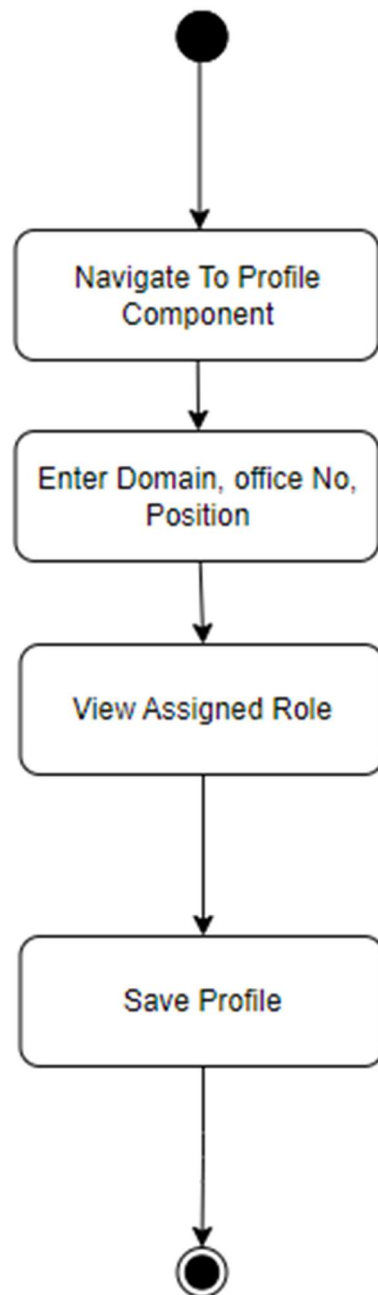


Figure 29 Activity Diagram Faculty Profile

4.6 Sequence Diagram

4.6.1 Login

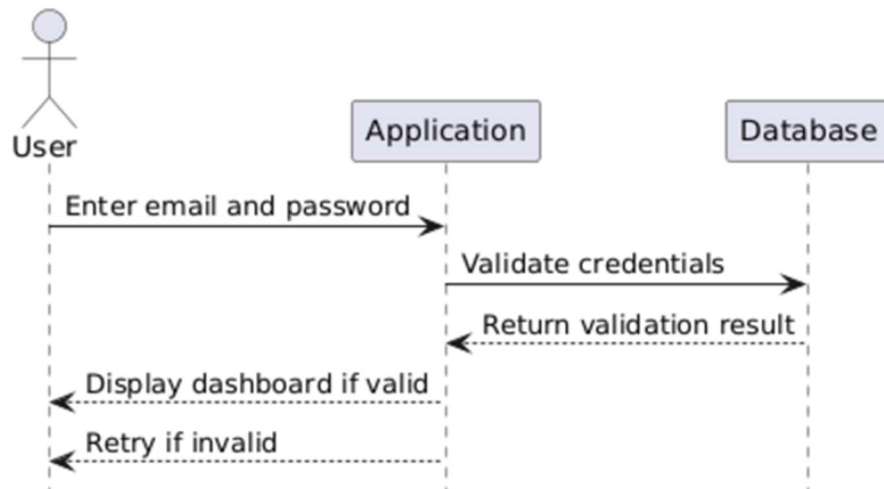


Figure 30 Sequence Diagram Login

4.6.2 Sign up

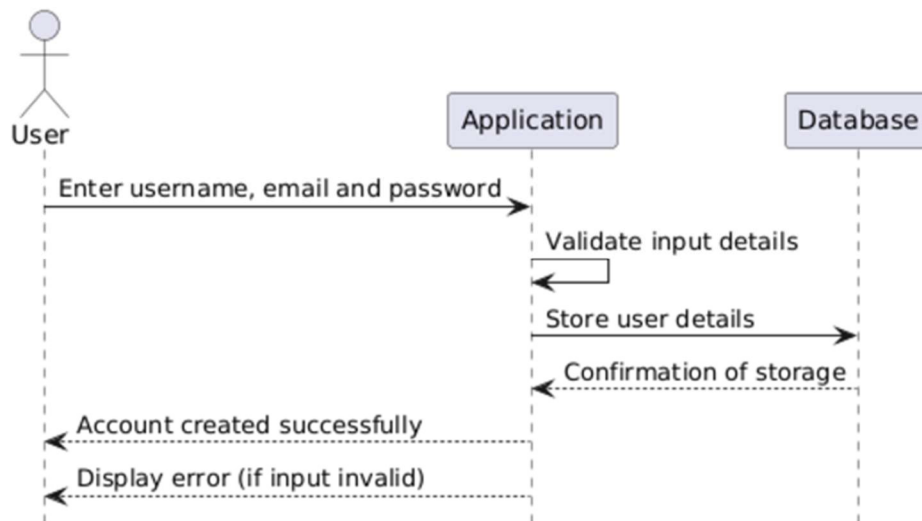


Figure 31 Sequence Diagram Sign Up

4.6.3 Log out

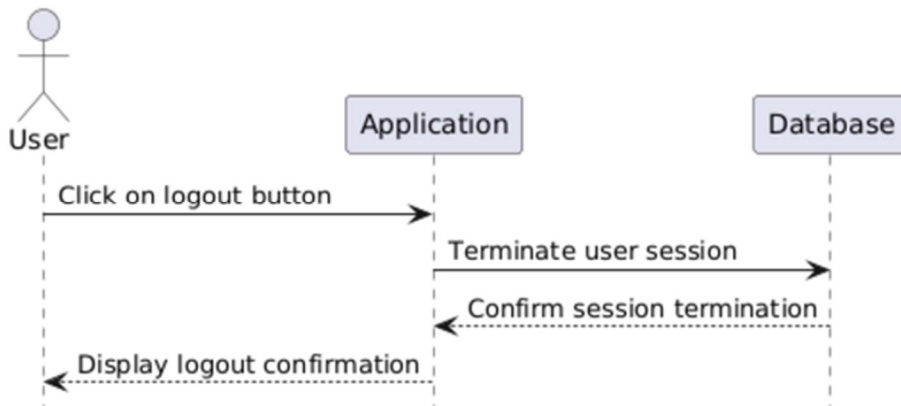


Figure 32 Sequence Diagram Logout

4.6.4 Student Request To Supervisor

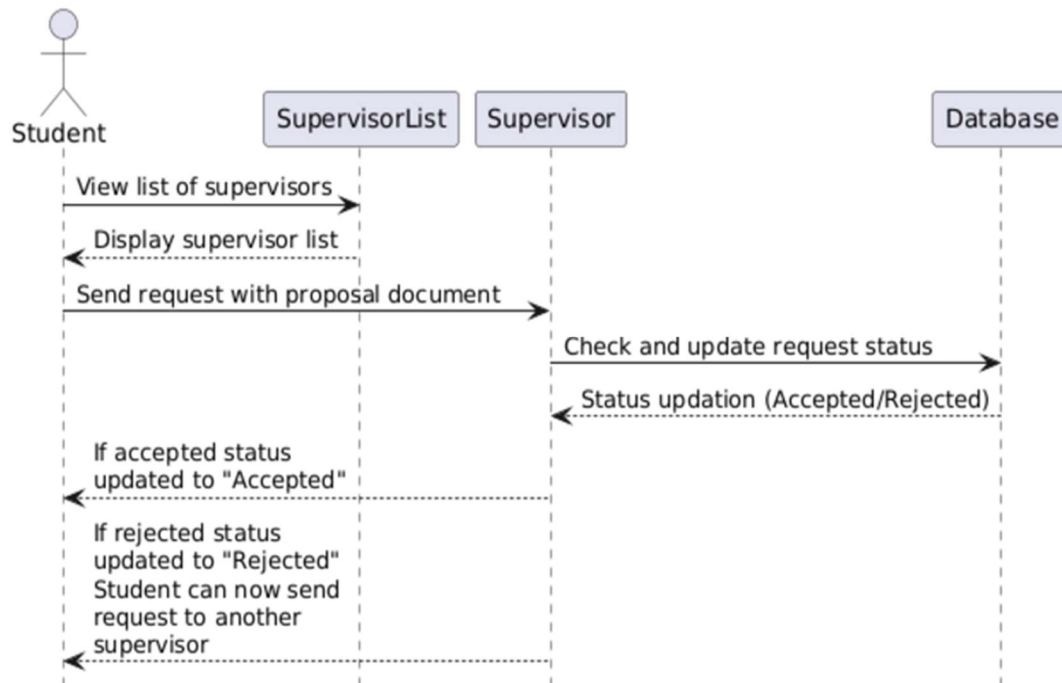


Figure 33 Sequence Diagram Request To Supervisor

4.6.5 Discussion Forum

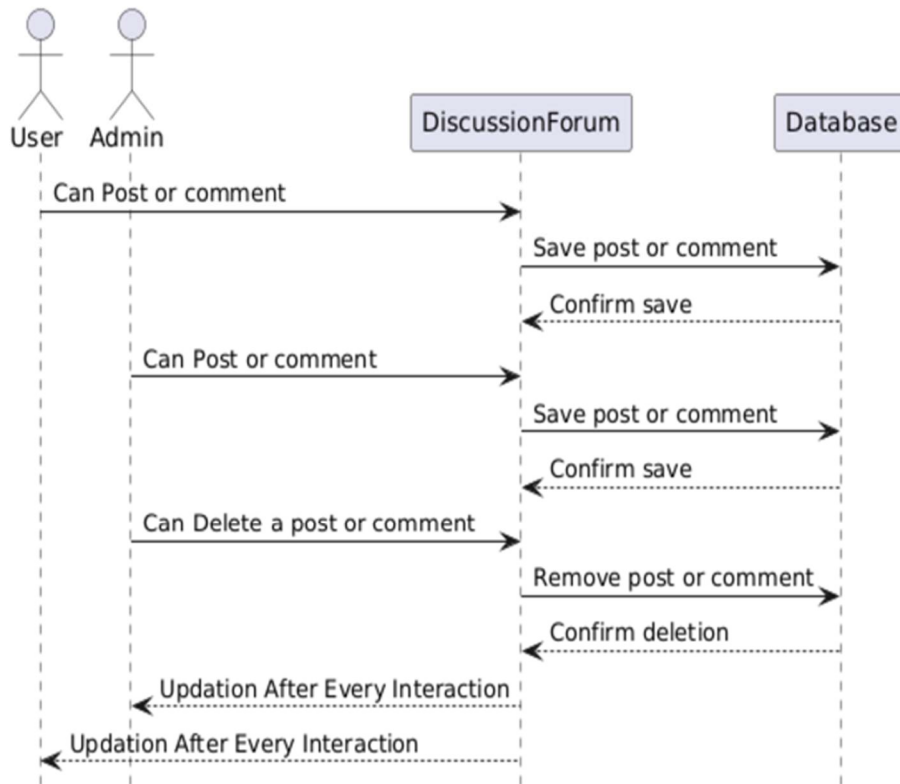


Figure 34 Sequence Diagram Discussion Forum

4.6.6 Announcement Section

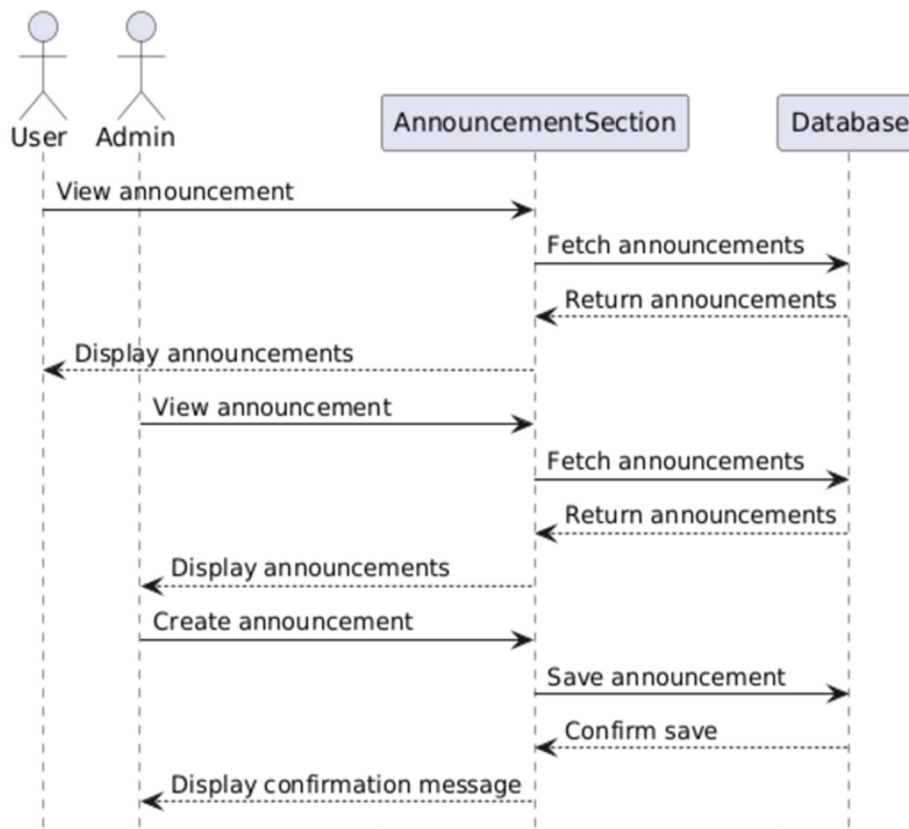


Figure 35 Sequence Diagram Announcement Section

4.6.7 Template

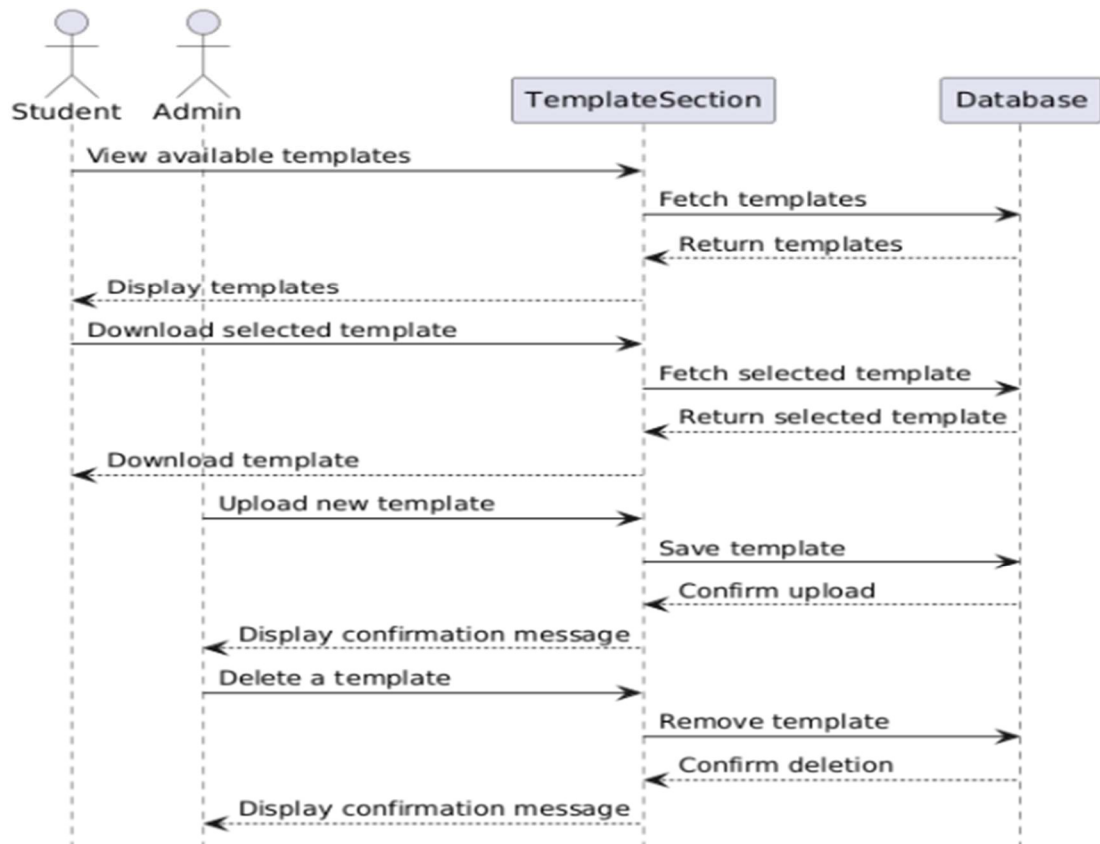


Figure 36 Sequence Diagram Template

4.6.8 Document Submission

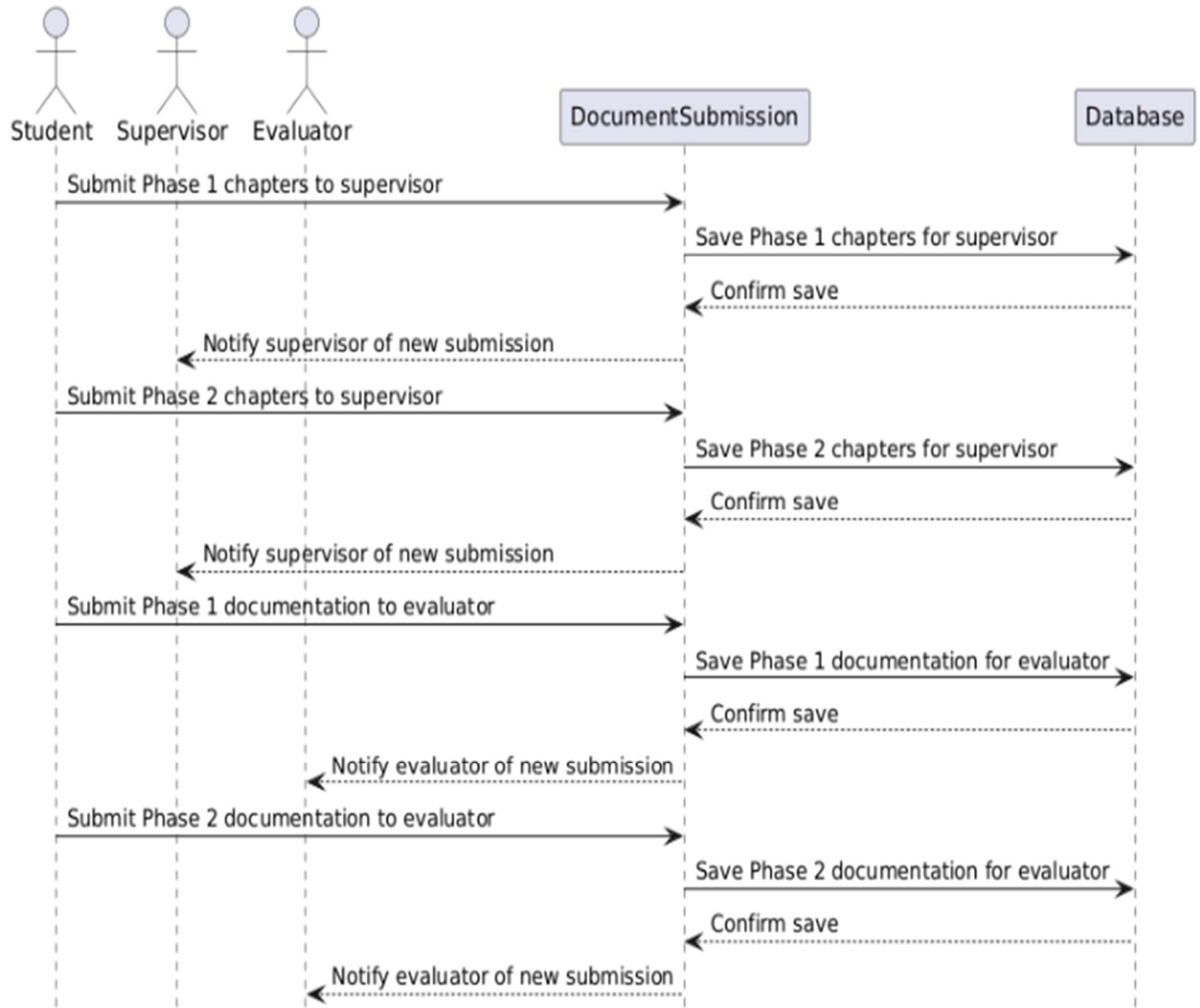


Figure 37 Sequence Diagram Document Submission

4.6.9 Chapters Evaluation

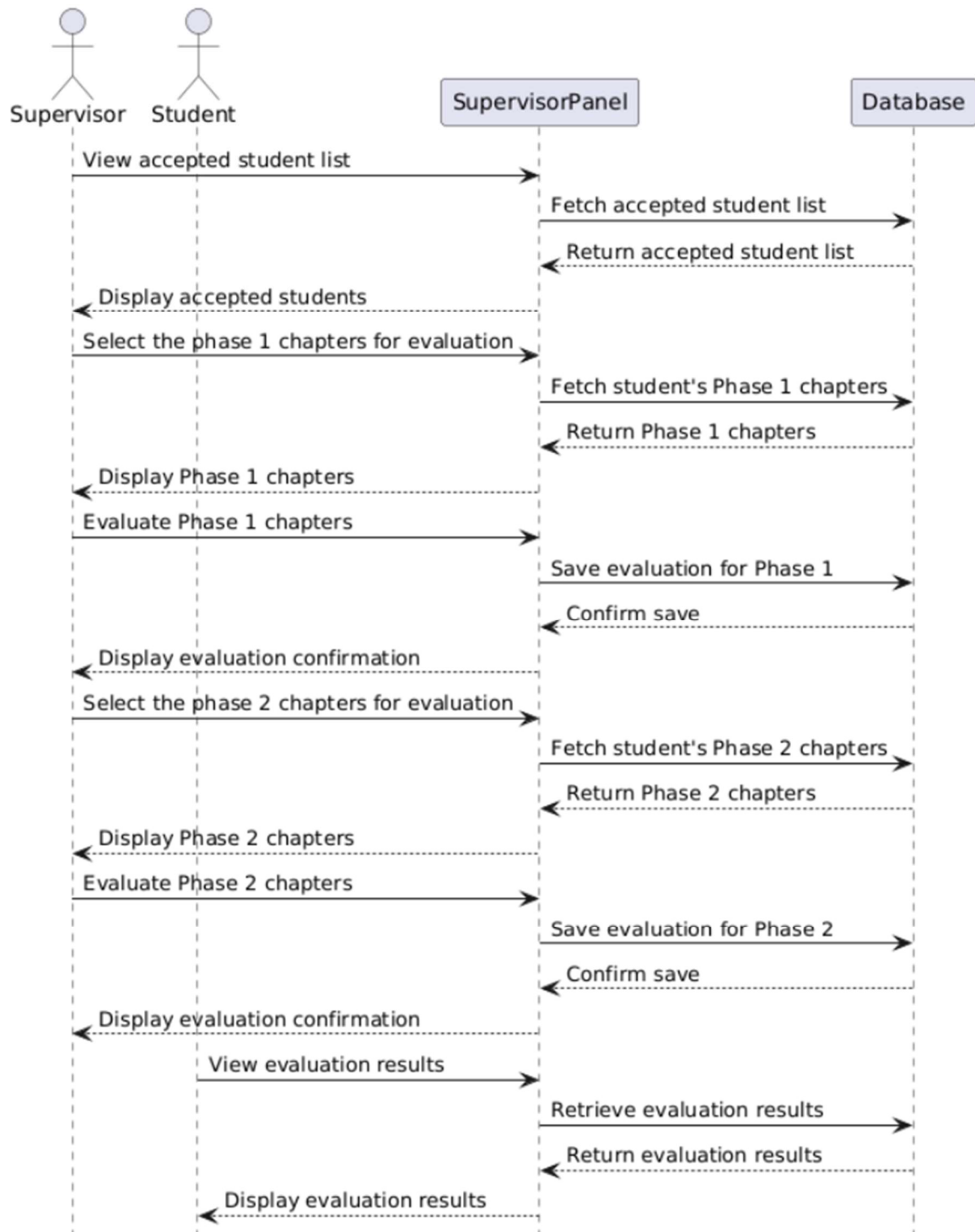


Figure 38 Sequence Diagram Chapters Evaluation

4.6.10 Documentation Evaluation

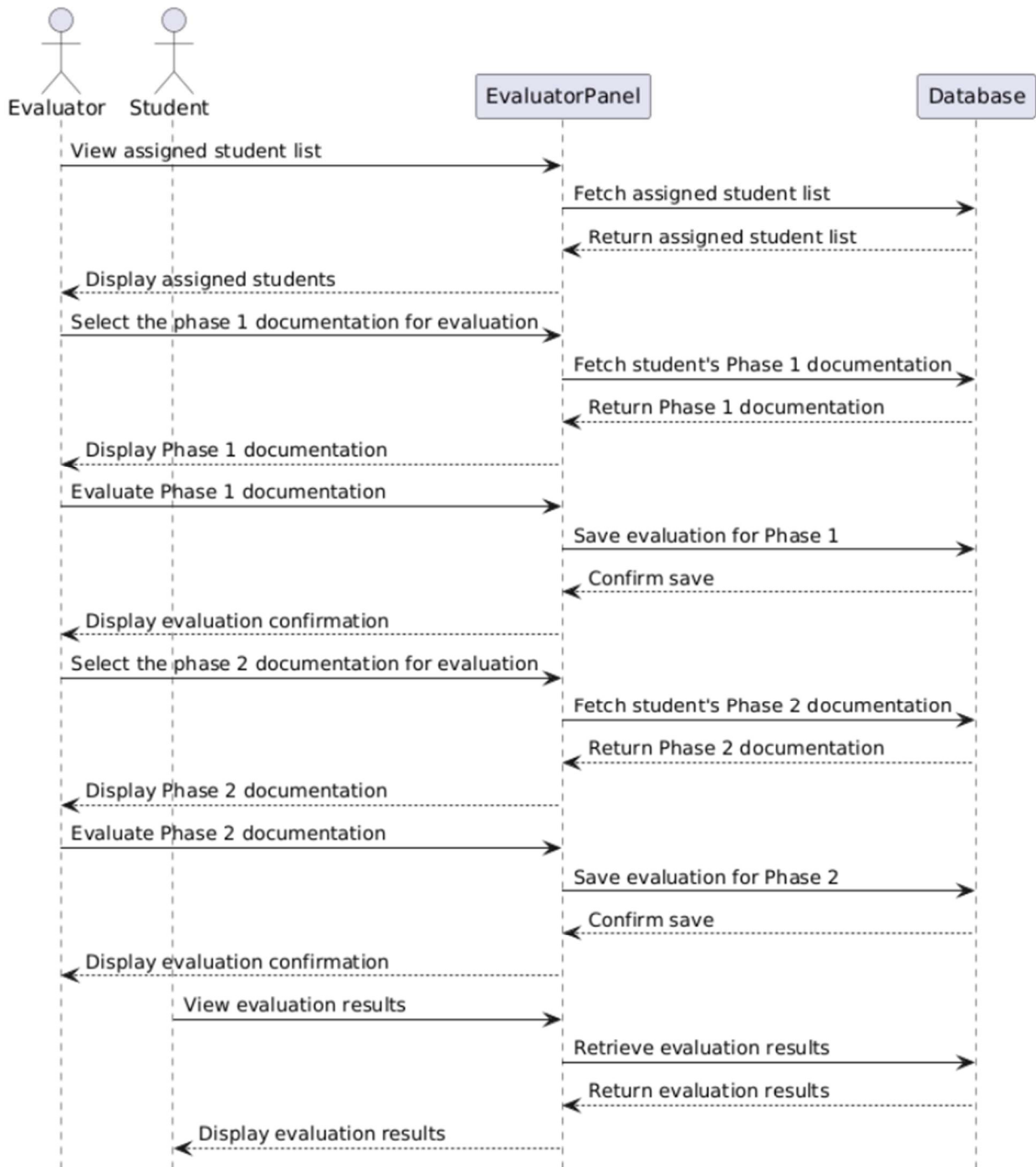


Figure 39 Sequence Diagram Document Evaluation

4.6.11 Supervisors, Evaluator, Projects List Creation

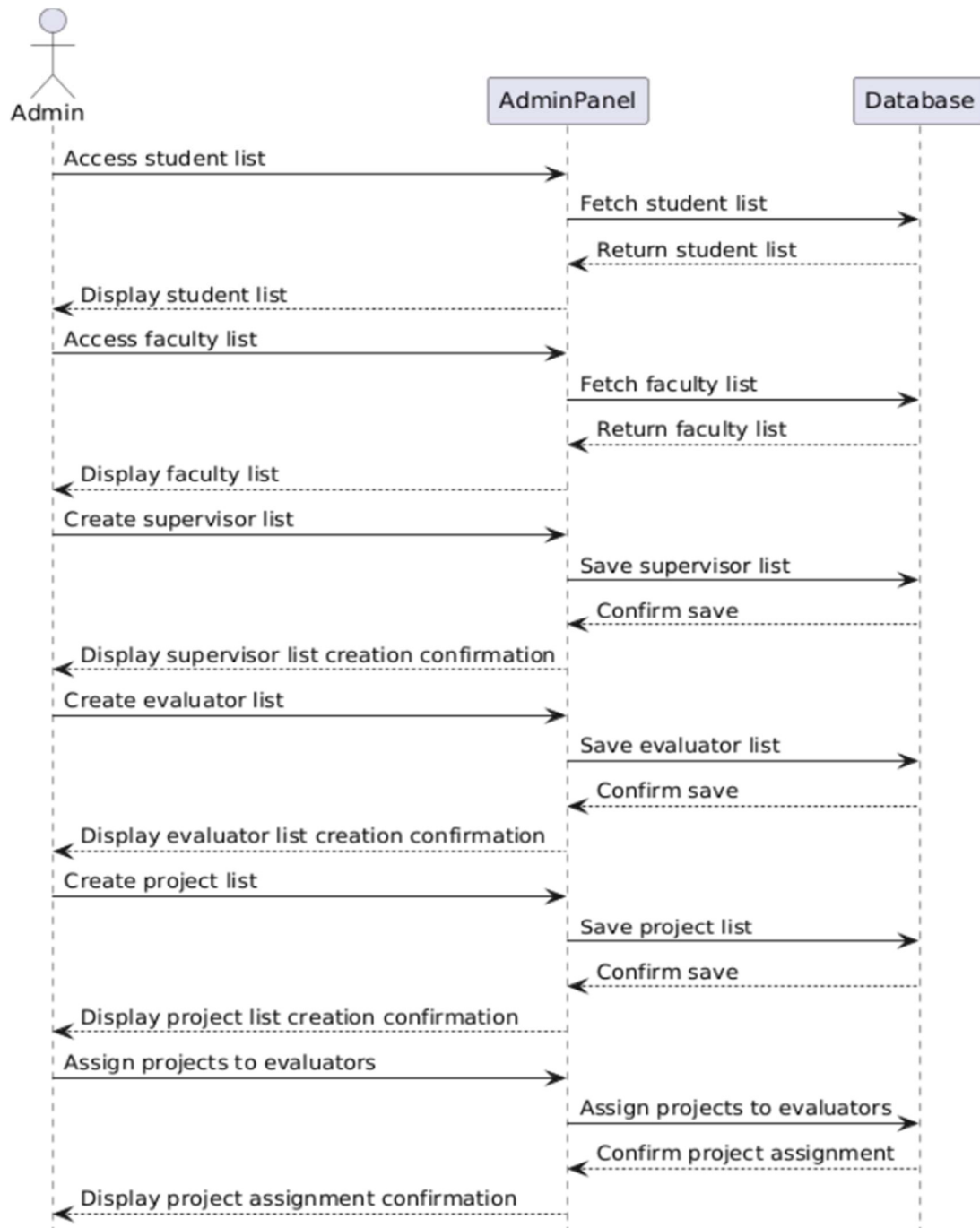


Figure 40 Sequence Diagram Supervisors, Evaluators, Projects List Creation

4.7 Collaboration Diagram

4.7.1 Login

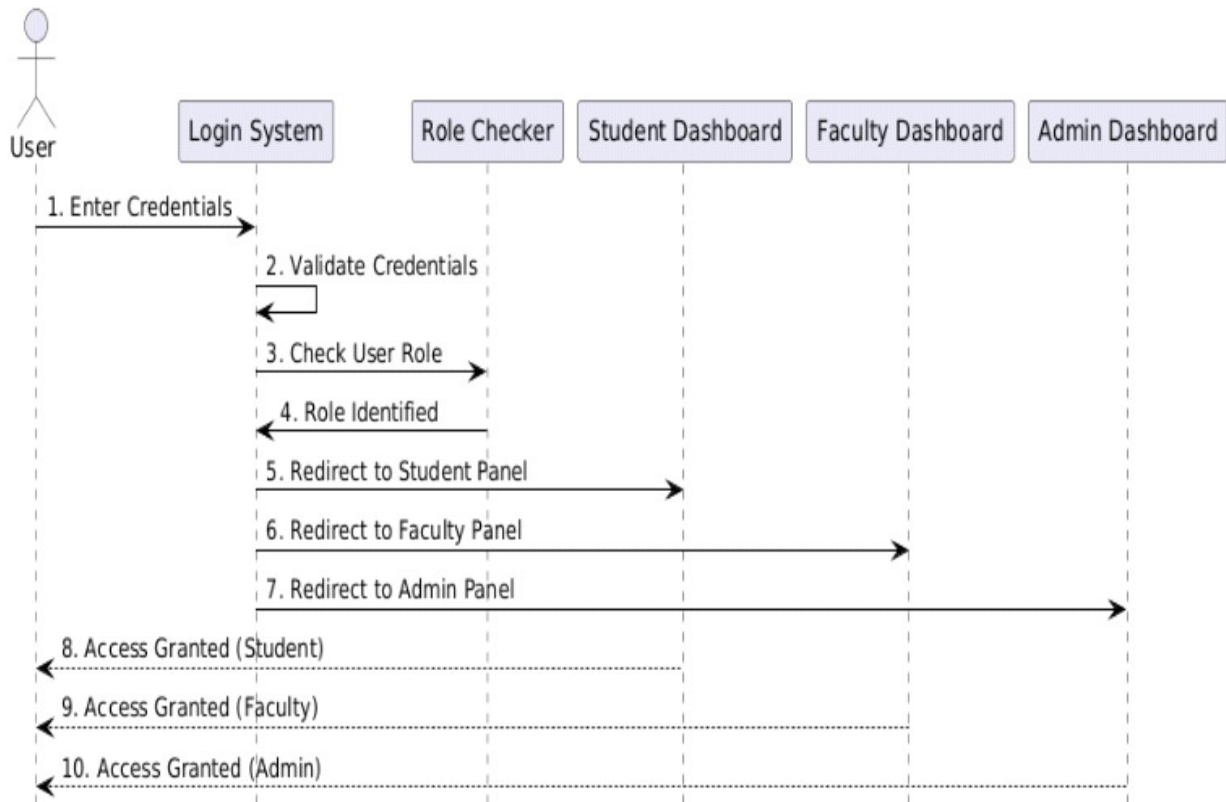


Figure 41 Collaboration Diagram Login

4.7.2 Student Panel

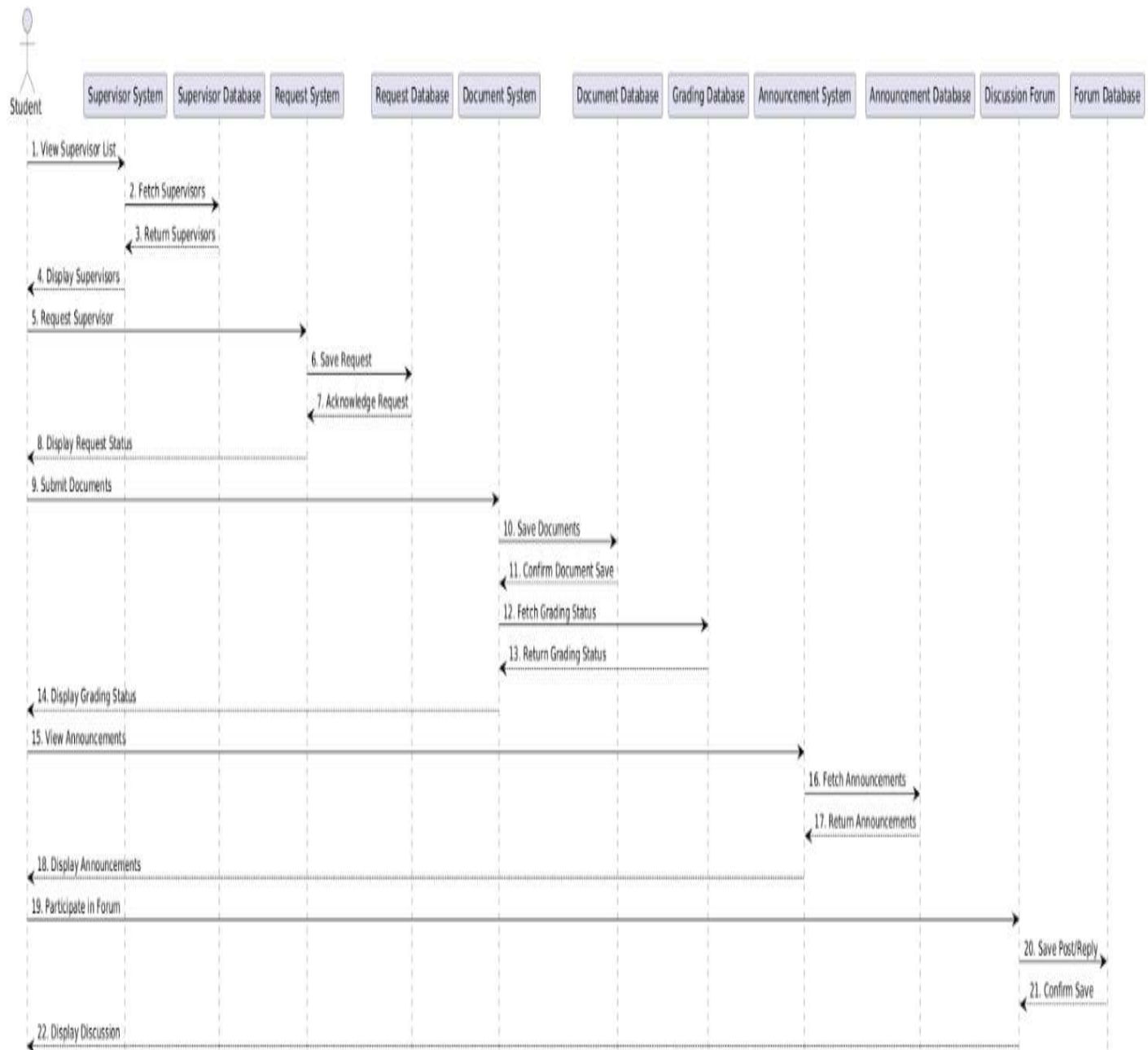


Figure 42 Collaboration Diagram Student Panel

4.7.3 Faculty Panel

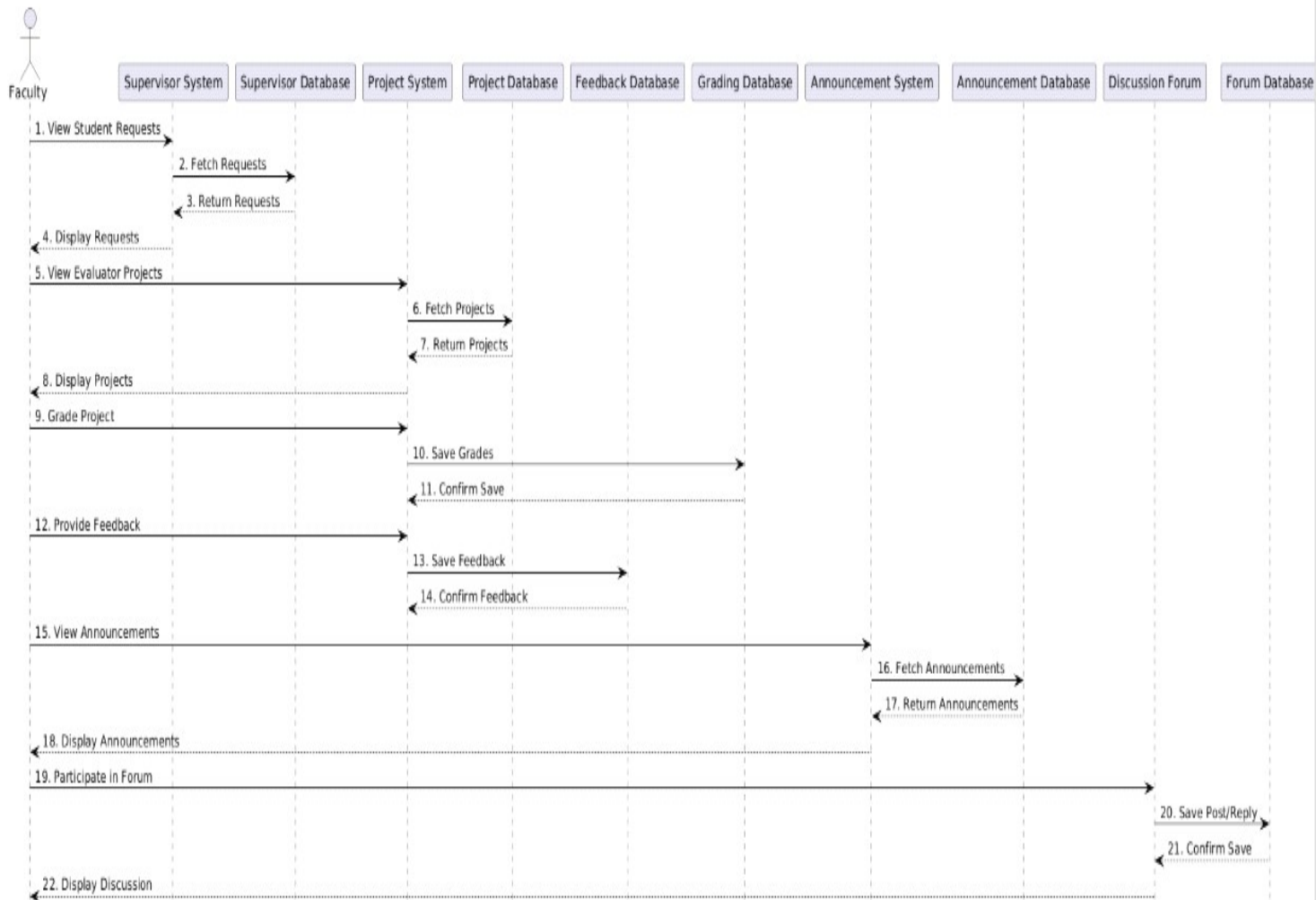


Figure 43 Collaboration Diagram Faculty Panel

4.7.4 Admin Panel



Figure 44 Collaboration Diagram Admin Panel

4.8 State Transition Diagram

4.8.1 Admin Panel

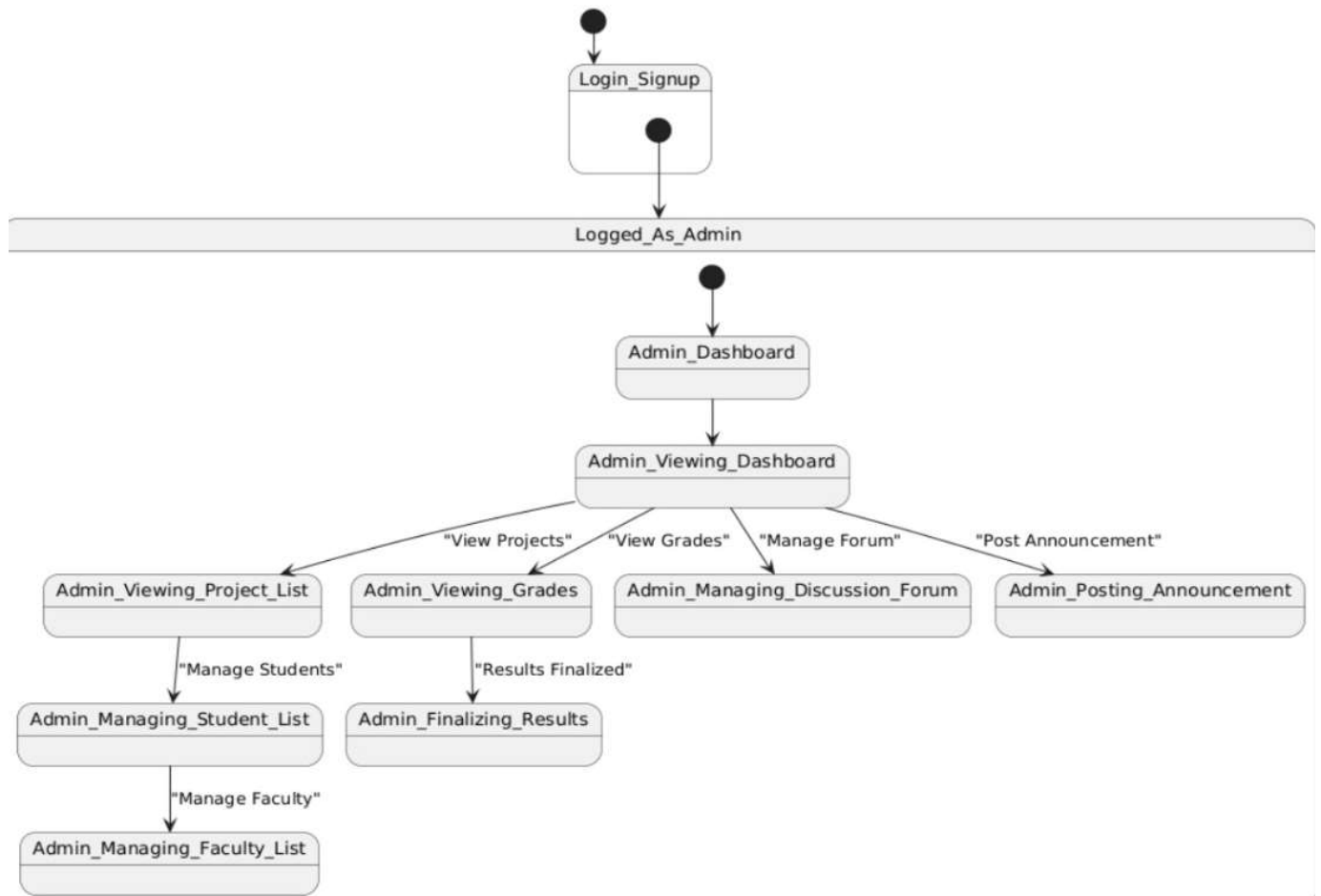


Figure 45 State Transition Diagram Admin Panel

4.8.2 Faculty Panel

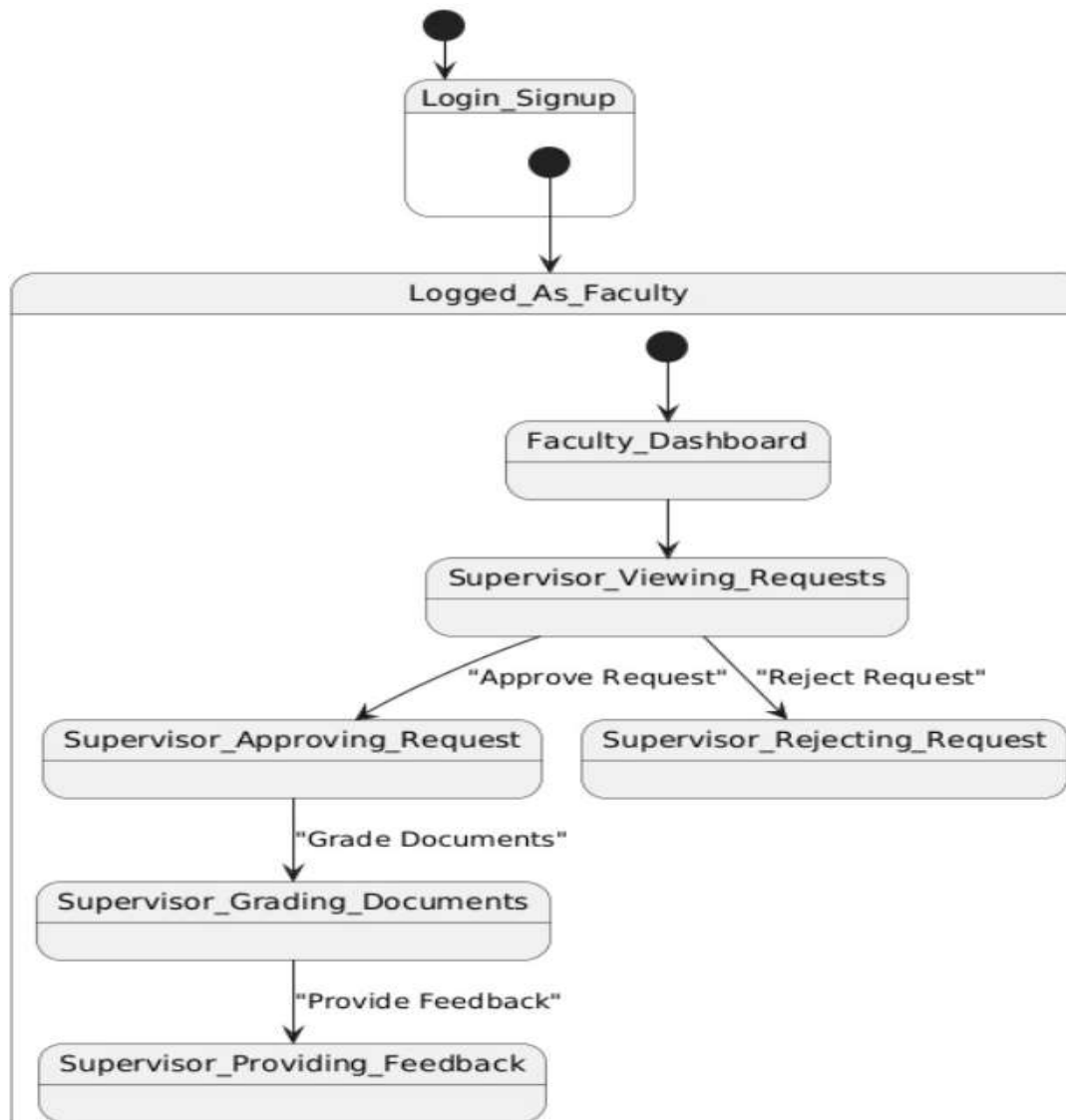


Figure 46 State Transition Diagram Faculty Panel

4.8.3 Student Panel

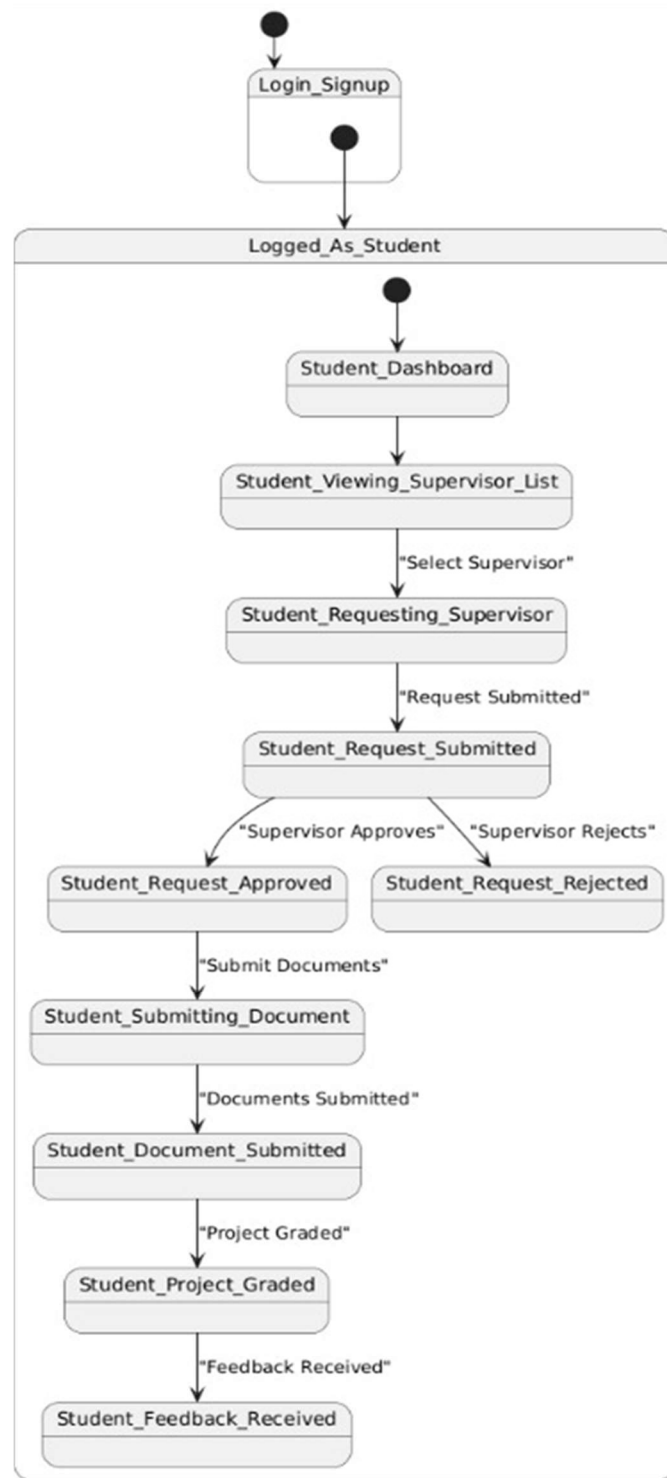


Figure 47 State Transition Diagram Student Panel

4.9 Component Diagram

4.9.1 Student Panel:

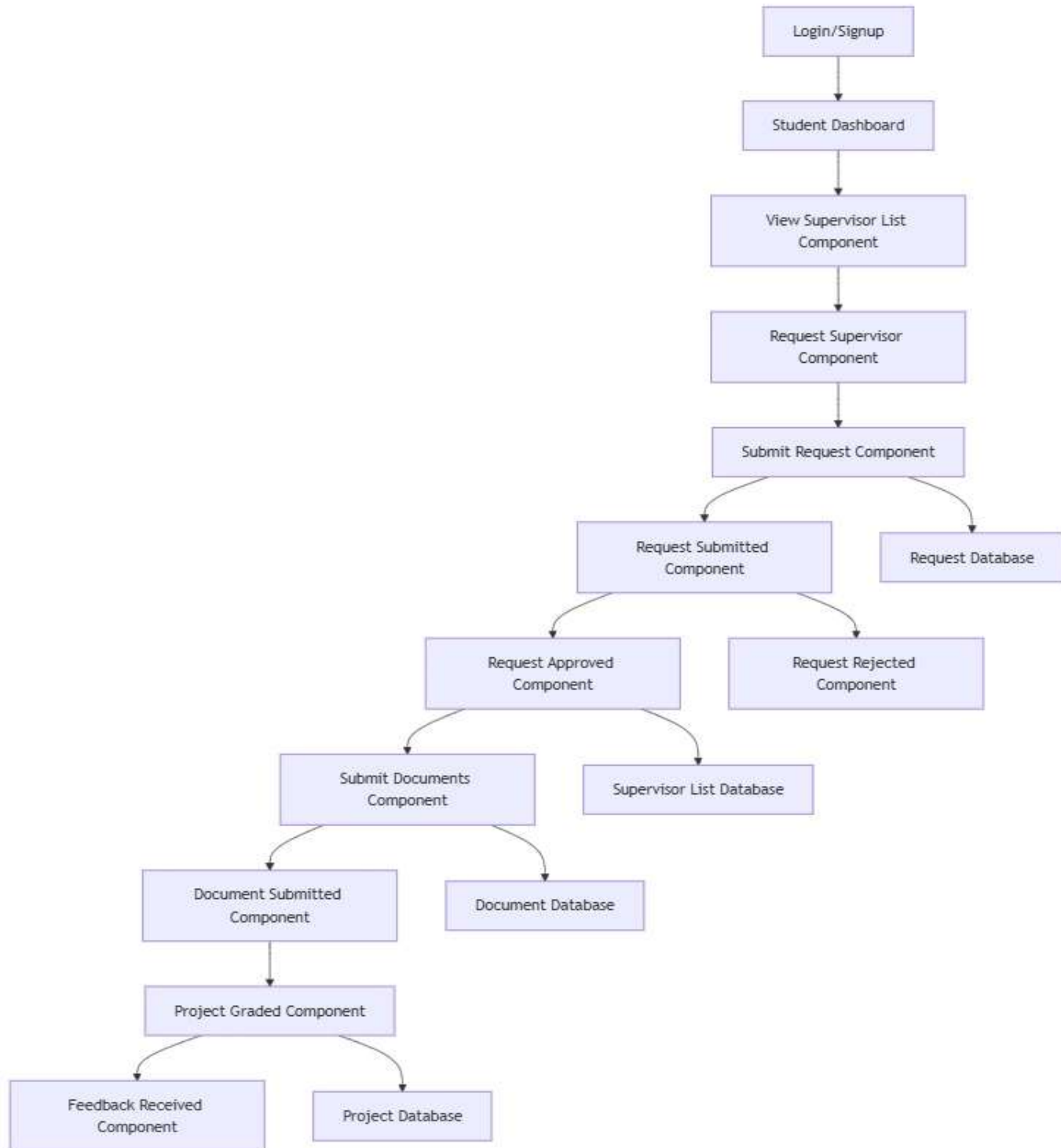


Figure 48 Component Diagram Student Panel

4.9.2 Faculty Panel

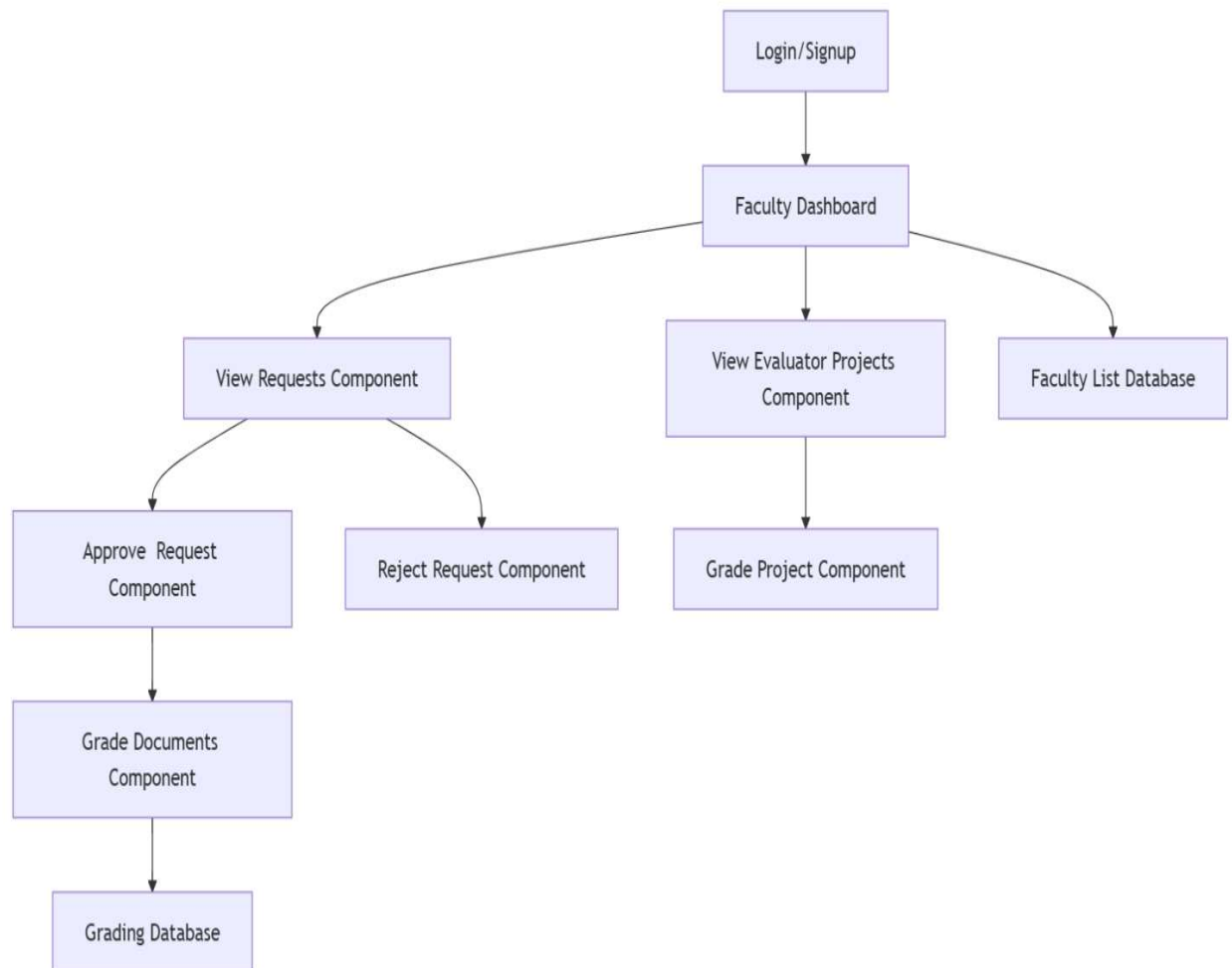


Figure 49 Component Diagram Faculty Panel

4.9.3 Admin Panel

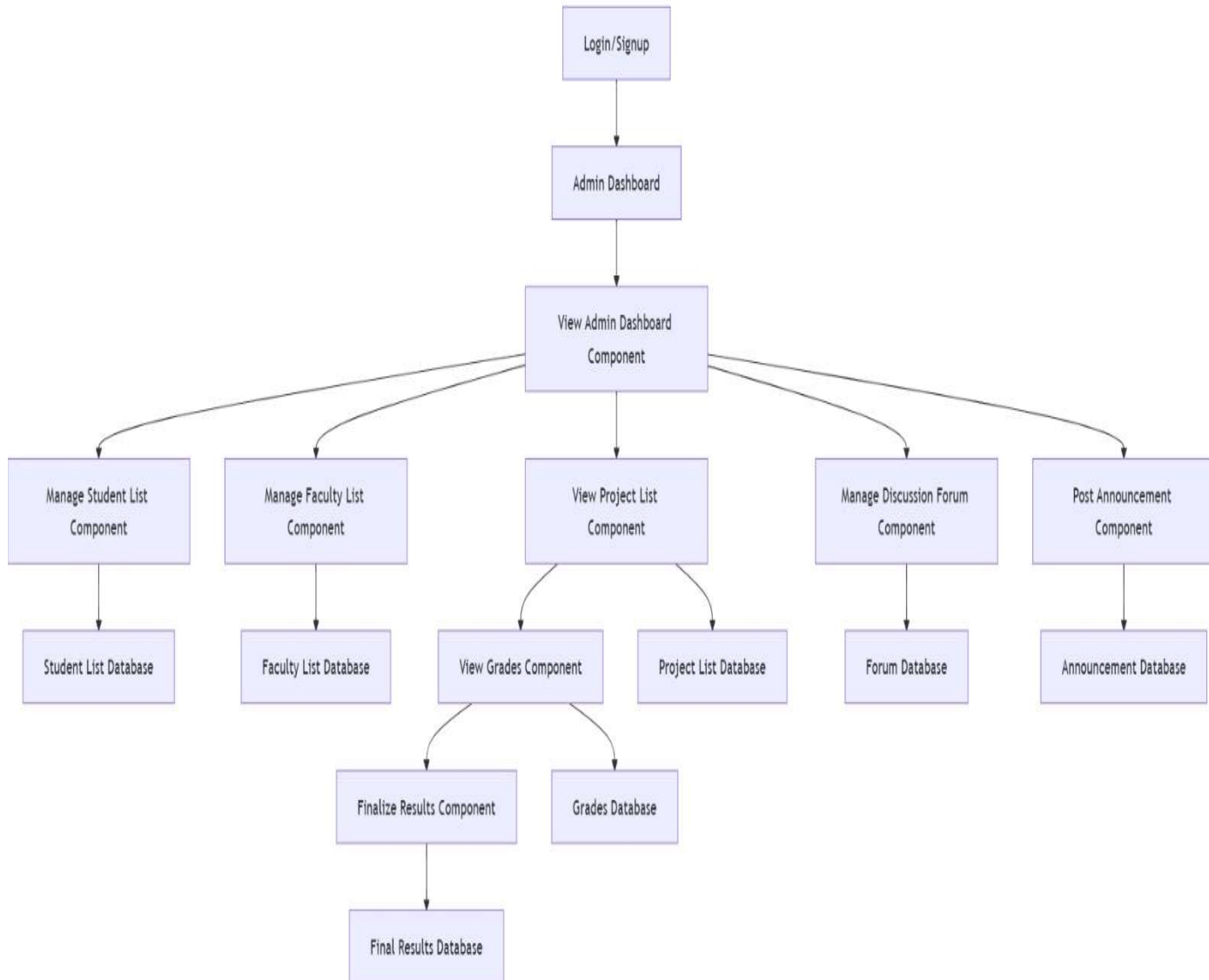


Figure 50 Component Diagram Admin Panel

4.10 Deployment Diagram

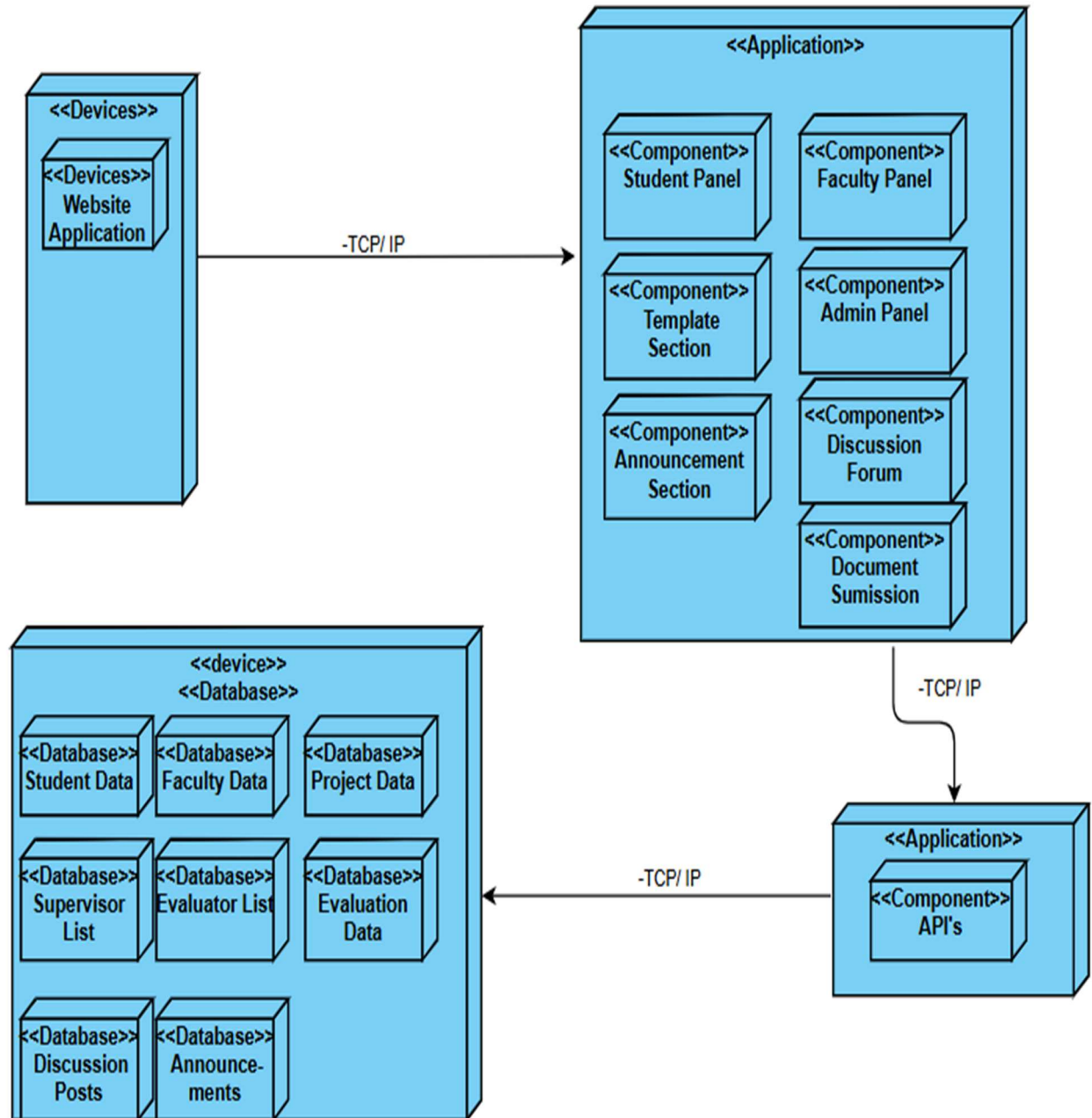
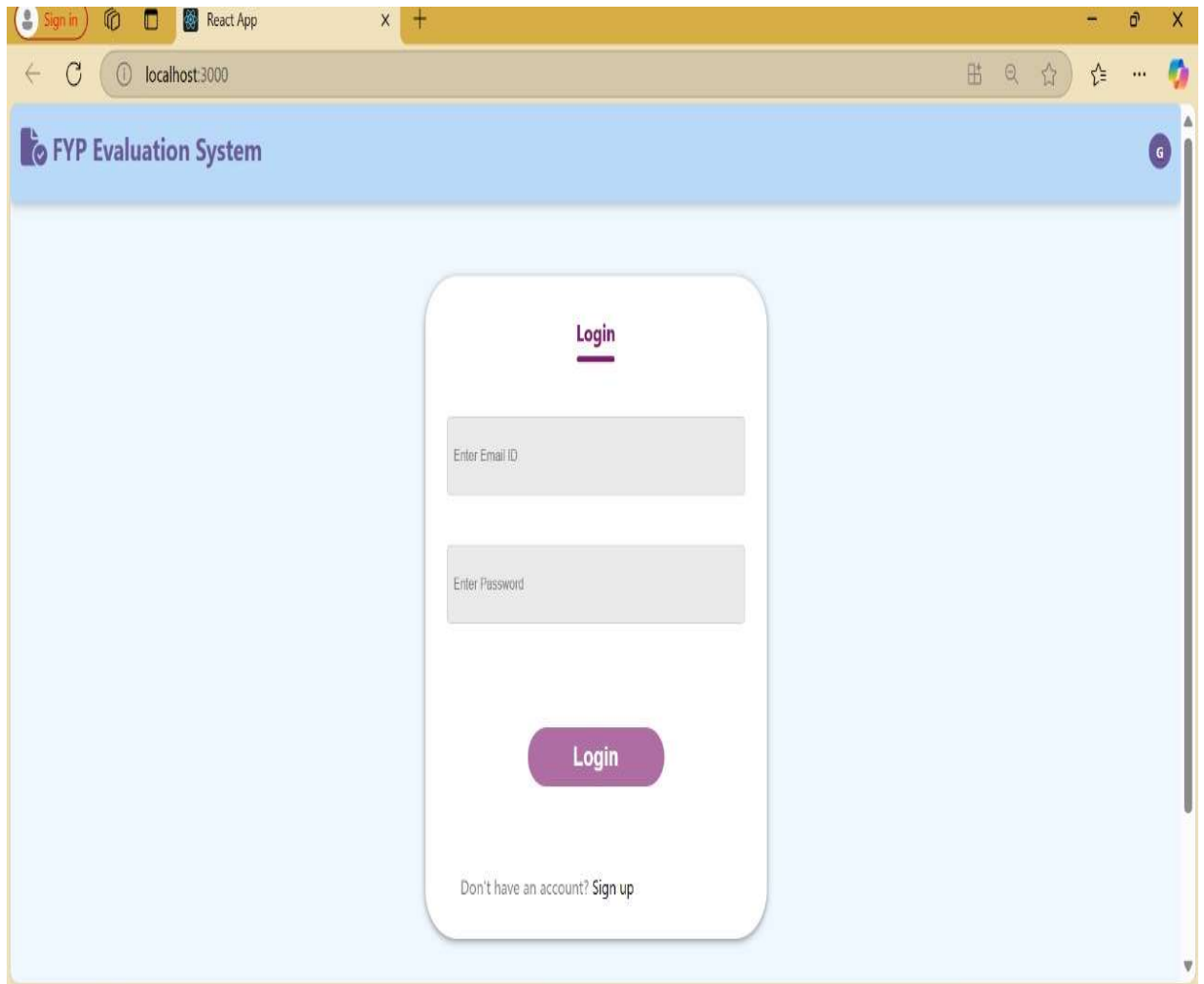


Figure 51 Deployment Diagram

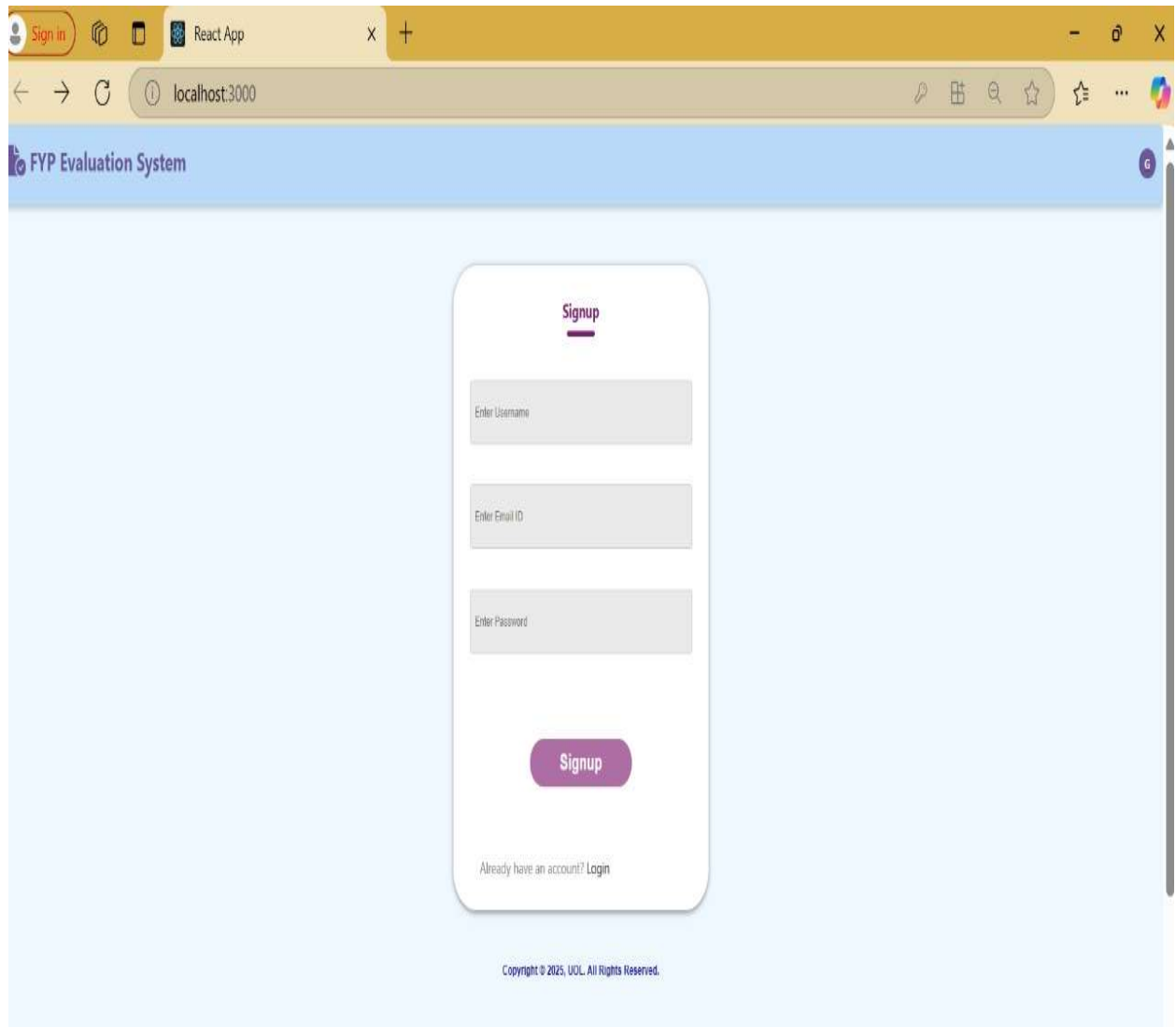
Chapter 5: User Manual

5.1 Login



The screenshot shows a web browser window with the address bar displaying 'localhost:3000'. The page title is 'FYP Evaluation System'. The main content area features a light blue background with a white rounded rectangle in the center. Inside this rectangle, the word 'Login' is displayed in purple text with a purple underline. Below this, there are two input fields: 'Enter Email ID' and 'Enter Password', both with light gray borders. A purple 'Login' button is positioned below the password field. At the bottom of the white rectangle, the text 'Don't have an account? Sign up' is displayed, with 'Sign up' as a link. The browser's top bar shows a 'Sign in' button and a 'React App' tab.

5.2 Signup



The screenshot shows a web browser window with a single tab titled "React App". The address bar displays "localhost:3000". The page header is a light blue bar with the text "FYP Evaluation System" on the left and a small purple circle with the number "6" on the right. The main content area has a light blue background. In the center, there is a white rounded rectangle with a purple shadow. At the top of this rectangle is the word "Signup" in purple, underlined. Below it are three input fields: "Enter Username", "Enter Email ID", and "Enter Password". Each field has a light gray border and a small purple icon on the right. Below the input fields is a purple rounded button with the text "Signup" in white. At the bottom of the rectangle, there is a link that says "Already have an account? Login". At the very bottom of the page, there is a small copyright notice: "Copyright © 2025, UOL. All Rights Reserved."

5.3 Discussion Forum

Discussion Forum
Welcome to t|

Post

Afifakaleembuti@student.uoL.edu.pk we finally won!! **Delete Post**

07/03/2024, 12:46:19 AM

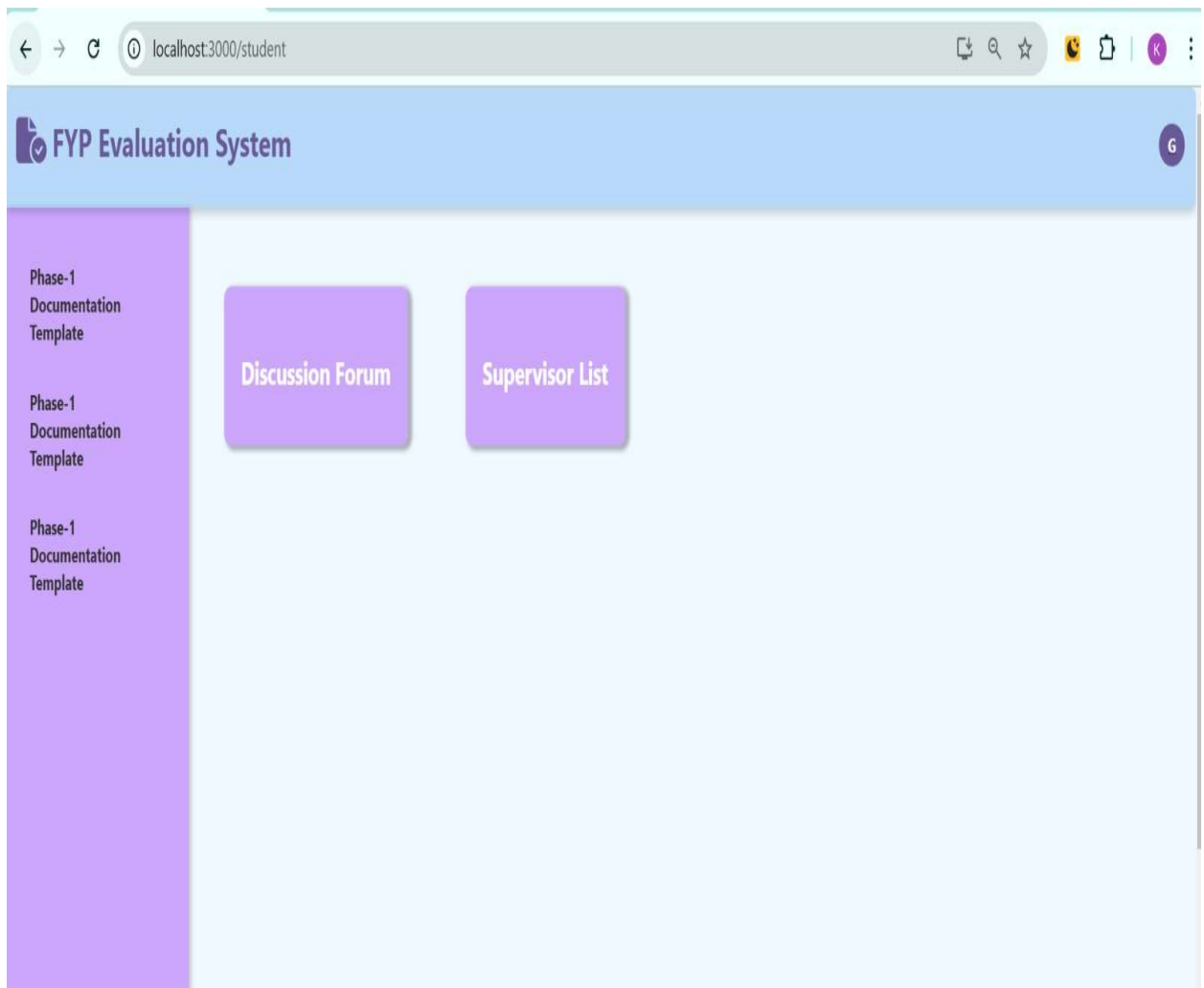
Reply to this post...

Reply

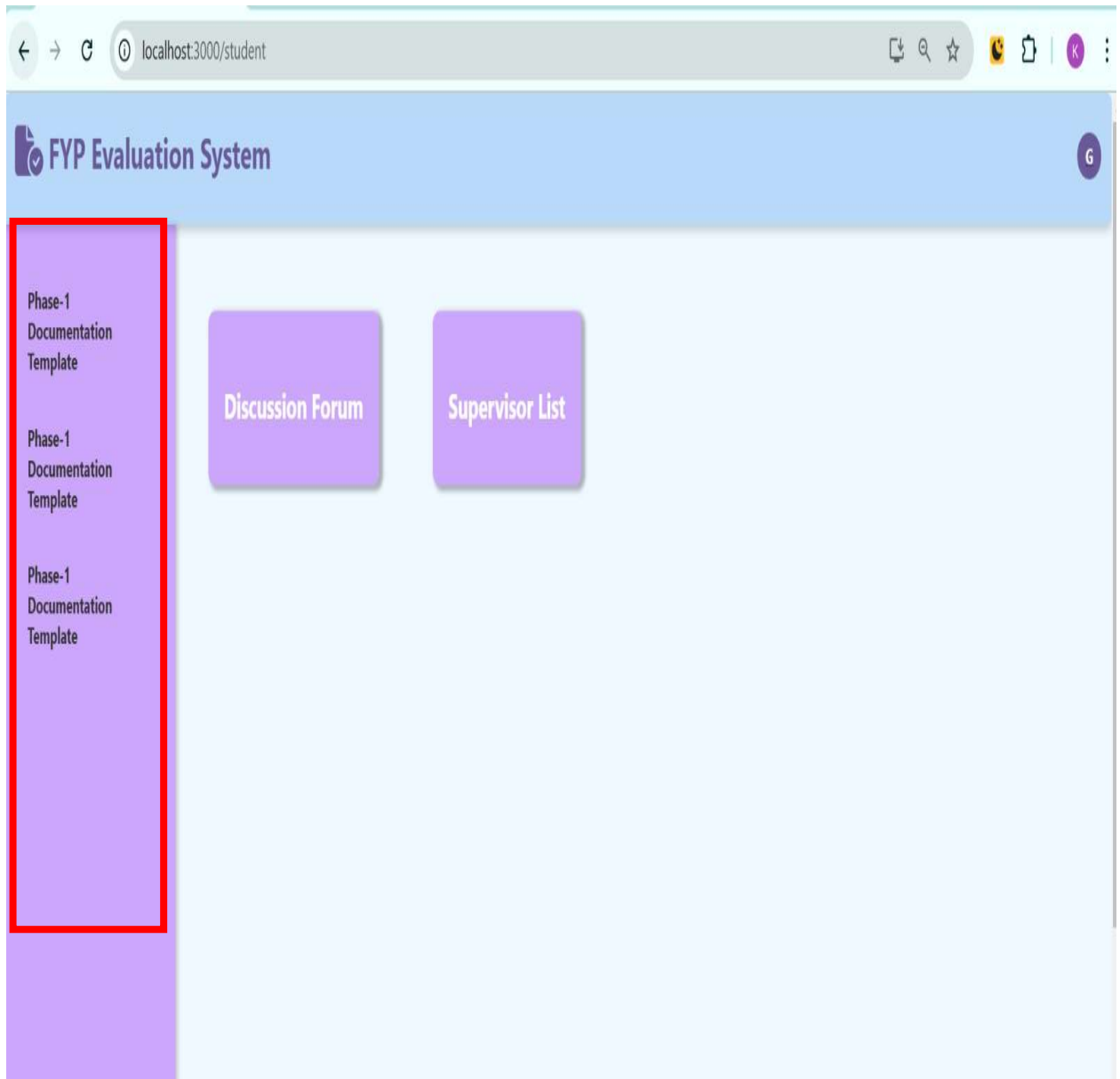
khazraasifkhan@student.uoL.edu.pk but it was hard! Replied at: 07/03/2024, 12:56:05 AM

Delete Reply

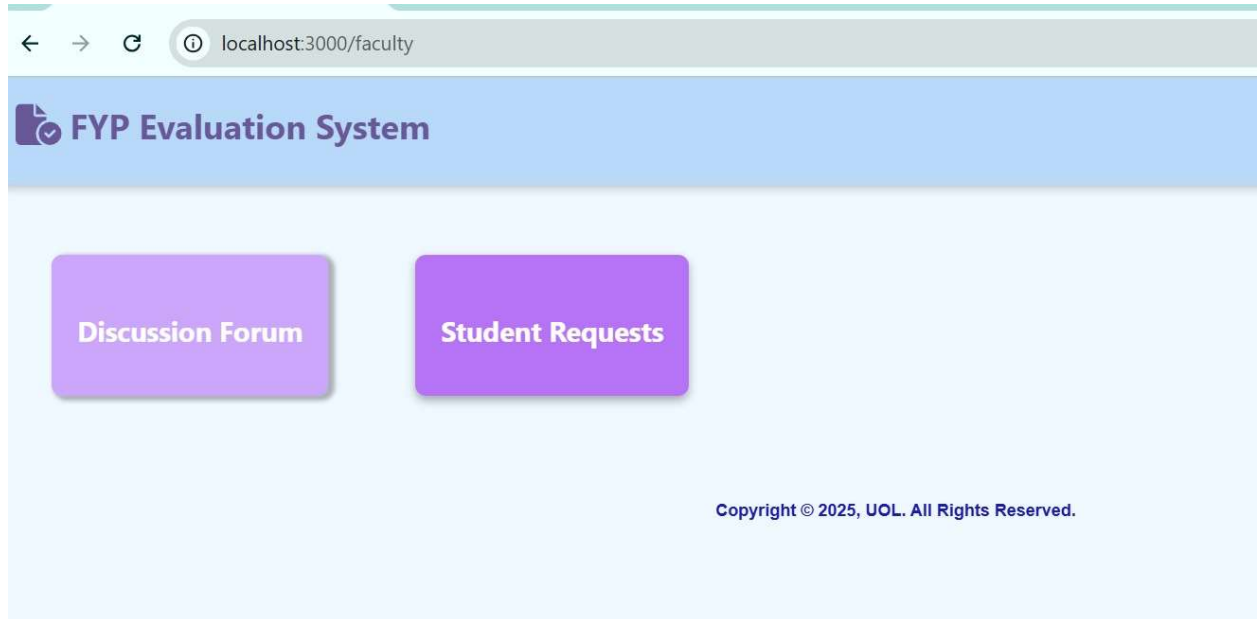
5.4 Student Panel:



5.5. Templates:

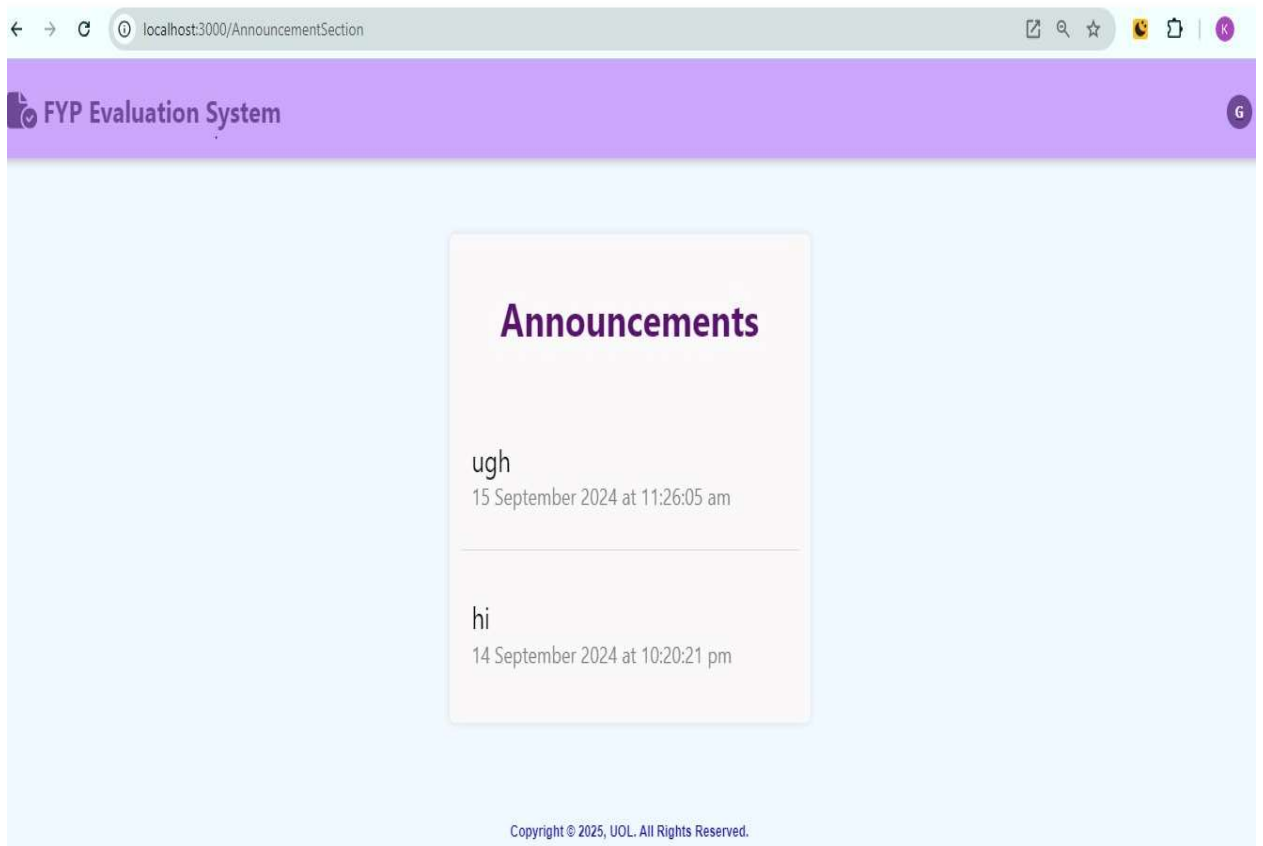


5.6. Faculty Panel:



5.7. Announcement Section:





References:

Book Reference

[1] I. Sommerville, Software Engineering, 10th ed. London: Pearson Education, 2016.

Journal Article Reference

[2] R. Pressman and B. Maxim, "Modern Practices in Software Project Evaluation," Journal of Software Engineering, vol. 45, no. 3, pp. 220-235, 2020.

Conference Reference

[3] A. Hussain, "Innovative Approaches to Final Year Project Management," in Proceedings of the International Conference on Educational Technologies (ICET), Lahore, Pakistan, 2024, pp. 105-117.

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[6] The University of Lahore, "Final Year Project Evaluation System User Manual," CS & IT Department, UOL, 2024 [Online]. Available: [<https://uol.edu.pk/fyp-system>].

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[7] React.js Documentation, "React Official Documentation," [Online]. Available:
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[8] Node.js Documentation, "Node.js Official Documentation," [Online]. Available:
[<https://nodejs.org>]. Accessed: Jan. 5, 2025.

Appendix:

The appendix provides supplementary information relevant to the **Final Year Project Evaluation System** but too detailed for inclusion in the main document. It ensures the main text remains concise and focused while offering additional resources for further clarity.

Appendix A: Acronyms and Abbreviations

This section defines the commonly used acronyms and abbreviations throughout the document.

Acronym/Abbreviation	Full Form
FYP	Final Year Project
FYPES	Final Year Project Evaluation System
UOL	The University of Lahore
SRS	Software Requirements Specification
API	Application Programming Interface
ERD	Entity Relationship Diagram
DFD	Data Flow Diagram
RAM	Random Access Memory
HTTP/HTTPS	Hypertext Transfer Protocol/Secure

Appendix B: User Roles and Responsibilities

The system defines different user roles, each with specific responsibilities:

Role	Responsibilities
Student	Upload project reports, interact on the discussion forum, and track evaluation progress.
Supervisor	Guide students, review and grade project documents, and provide feedback.
Evaluator	Assess project documentation and assign grades.
Admin	Manage user roles, announcements, templates, and system configurations.

Appendix C: Functional Requirements

A list of key functional requirements that the system fulfills:

Requirement ID	Description
FR_01	Users can create an account (Signup).
FR_02	Users can log in to their accounts.
FR_16	Students can upload reports for evaluation.
FR_17	Faculty members can grade submitted reports.

Appendix D: Use Case Descriptions

The use cases describe interactions between users and the system.

Use Case ID	Use Case Name	Description
UC_01	Login	Users log into the system to access their portal.
UC_07	Request to Supervisor	Students send requests to supervisors for project guidance.
UC_08	Document Submission	Students submit their documents or chapters for evaluation.
UC_12	Document Evaluation	Evaluators review and provide feedback on submitted documents.

Appendix E: Design Diagrams

The system's design is represented using various diagrams to provide a clear understanding of the architecture and workflow.

Architecture Diagram – Visual representation of the system's components and their interactions.

Entity Relationship Diagram (ERD) – Shows the relationships between entities within the system.

Data Flow Diagram (DFD) – Depicts how data flows through the system.

Class Diagram – Displays the structure of the system in terms of classes and their relationships.

Activity Diagram – Describes workflows of various system processes.

Appendix F: Evaluation Criteria

The system allows supervisors and evaluators to assess student projects based on predefined criteria.

Evaluation Criteria	Description
Proposal Evaluation	Assessment of the project proposal submitted by the student.
Progress Report Grading	Grading of progress reports at different project phases.
Final Documentation	Evaluation of the final project documentation.
Presentation Grading	Assessment of the final presentation delivered by the student.

Appendix G: System Screenshots

Screenshots of key system interfaces, such as the login screen, student dashboard, faculty dashboard, and admin panel, are provided to help visualize the user experience.

1. Login Page
2. Student Panel
3. Discussion Forum
4. Admin Panel
5. Announcement Section
6. Templates

Appendix H: System Requirements

The system requirements to run the Final Year Project Evaluation System include hardware and software specifications.

Hardware Requirements

Server: Minimum 8 GB RAM, 256 GB SSD, Quad-Core Processor.

Client Devices: Desktop, laptop, or mobile device with a web browser.

Software Requirements

Operating System: Windows

Browser: Google Chrome, Mozilla Firefox

Backend: Node.js

Frontend: React.js

Appendix I: Discussion Forum Guidelines

The discussion forum is a key feature of the system, designed to enhance communication between students and faculty.

Guidelines for Use:

Students must adhere to respectful communication standards.

Supervisors and evaluators should respond to queries within a reasonable timeframe.

Spam or irrelevant posts will be removed by the admin.